Elimination Rules Playbook

Evidence-based last-resort heuristics for Civil, Electrical, and Non-core MCQs.

Publication floor: each established rule has at least 15 independent supporting questions from FLTs, Subject Tests, and VALID PYQs. Target healthy band: 15–20+ examples. **Diagram evidence is shown as actual figures** (not filename titles).

How to apply these rules (exam workflow)

1. **Solve normally first** — formula, figure, code clause.
2. If stuck between two or three options, **classify the trap family** (geometry, pu base, sequence fault, service vs factored, etc.).
3. Open the matching rule below and run the **Application steps**.
4. **Strike** options that violate the rule; keep only survivors.
5. If two remain, use Rule 17 (units / figure consistency). Never use letter-position myths (pick C, longest option, middle value).

Drill practice: 50-question jumbled tests are in the same folder (Test 01 covers Rules 1–5, Test 02 Rules 6–10, Test 03 Rules 11–15, Test 04 Rules 16–17).

1. Established heuristics — index

#	Rule	Streams	Evidence
1	Diameter ↔ radius / area geometry slip	Civil, Electrical	129
2	Service vs factored / partial-factor mix	Civil	104
3	Gross vs net / hole deduction	Civil	95
4	Cover vs effective depth	Civil	40
5	Per-unit base-change scaling	Electrical	95
6	Sequence-network connection by fault type	Electrical	25
7	√3 / phase-factor forgotten	Electrical	24
8	Thevenin / Norton source deactivation	Electrical	18
9	Resonance / $X_L = X_C$ identity	Electrical	47
10	Sign / sense / tension–compression flip	Civil, Electrical	270
11	Figure-dependency: cover the figure test	Civil, Electrical	304
12	Rankine / Terzaghi / earth-pressure term drop	Civil	86
13	Mohr / principal-stress pair consistency	Civil	26
14	SFD↔BMD jump / couple vs point-load confusion	Civil	39

#	Rule	Streams	Evidence
15	SCR / PE latching vs holding vs firing	Electrical	18
16	Non-core: option that abandons the asked operation	Non-core	37
17	Last-resort: dimensional / absurdity cull (evidence-limited)	Civil, Electrical, Non-core	232

Stream coverage

- **Civil:** R1, R2, R3, R4, R10, R11, R12, R13, R14, R17
- **Electrical:** R1, R5, R6, R7, R8, R9, R10, R11, R15, R17
- **Non-core:** R16, R17

All three streams are covered by the established set.

2. Rules — explanation and application

Elimination Rule 1 — Diameter ↔ radius / area geometry slip

Evidence: ESTABLISHED — 129 independent (125 bank + 4 VALID PYQ) · **Streams:** Civil, Electrical

What this rule means

Circular geometry questions almost always hide a diameter-versus-radius mistake. Area is A = pi*d^2/4. Candidates who treat the given diameter as a radius get areas four times too large (or stresses four times too small).

When to use

Circular section, shaft, hanger, pipe, or bolt with diameter given in mm.

Application steps

1. Circle the numeric diameter in the stem and write r = d/2 on scratch paper.
2. Compute A = pi*d^2/4 once; do not reuse a remembered 'pi r^2' with d plugged in as r.
3. Strike any option that matches P/(pi*d^2), P/(pi*r), or 4x/0.25x of the correct stress.
4. If net area of a plate with holes is asked, apply hole deductions after the correct gross area.

```
Memory cue: A = pi*d^2/4 ; sigma = P/A
```

Core elimination move

Compute $A = \pi d^2/4$ (or $r = d/2$) once on scratch paper. Eliminate options that match $P/(\pi d^2)$, $P/(\pi r)$, or treating d as r.

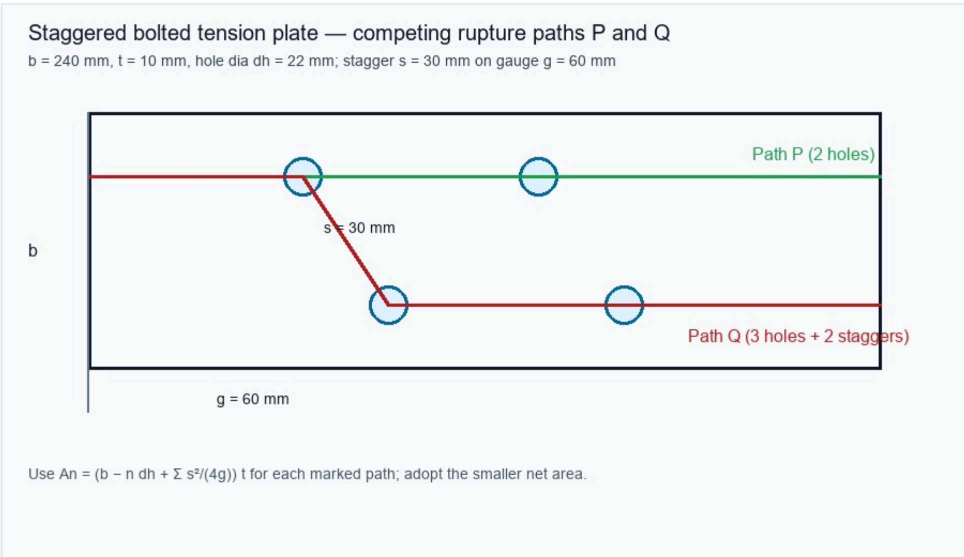
Evidence examples (sample references)

Source exam	Reference	Stem (truncated)
Civil FLT-01	`data/civil/ce-flt01.js#Q1`	Following a gantry-tie inspection at a substation, the engineer records a 20 mm diameter member under 55 kN...
Civil FLT-01	`data/civil/ce-flt01.js#Q2`	While reviewing a rigid-ended hanger, equal-length steel and brass rods are found to have \(A_s=600\) mm², ...
Civil FLT-01	`data/civil/ce-flt01.js#Q13`	During beam-section selection, the service moment is 36 kN·m and the allowable bending stress is 150 MPa. D...
Civil FLT-02	`data/civil/ce-flt02.js#Q1`	A 2.0 m long steel tie of uniform 500 mm² area carries a gradually applied 50 kN service load. Neglecting s...
Civil ST-FM	`data/civil/st/ce-st-fm-01.js#Q2`	Water flows in a 0.2 m diameter pipe at mean velocity 4 m/s. Discharge is closest to:
Civil ST-RCC	`data/civil/st/ce-st-rcc-01.js#Q1`	An RCC beam of M25 and Fe500 has factored Mu = 180 kN m, b = 230 mm and d = 450 mm. Using z = 0.9d, the req...

Figure evidence (actual diagrams — not path titles)

Civil FLT-01 · `data/civil/ce-flt01.js#Q47` · Key D

A bolted tension plate has width b = 240 mm, thickness t = 10 mm and hole diameter dh = 22 mm. Path P is a chain section through 2 holes; path Q zig-zags through 3 holes with two stagger credits s = 30 mm on gauge g = 60



A. P, 1960 mm²

B. P, 1740 mm²

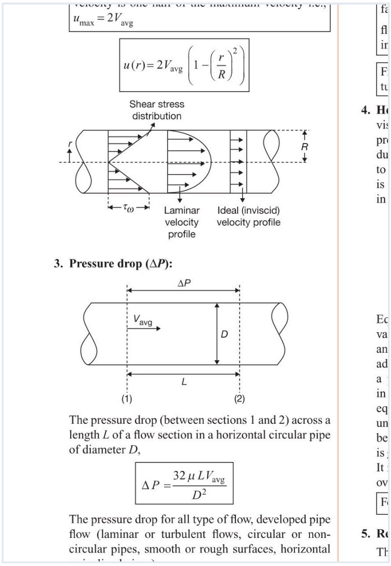
C. Q, 2030 mm²

D. Q, 1810 mm²

Embedded figure file: images/diagrams/civil-flt01/q47-staggered-net-paths.jpg

Civil FLT-02 · `data/civil/ce-flt02.js#Q35` · Key C

The pump and system curves show one pump and two identical pumps in parallel; the operating intersections are labelled only on the crop. Why is the parallel discharge at point C less than twice the single-pump discharge


$u_{max} = 2V_{avg}$

$$u(r) = 2V_{avg}\left(1 - \left(\frac{r}{R}\right)^2\right)$$

3. Pressure drop (ΔP):

The pressure drop (between sections 1 and 2) across a length L of a flow section in a horizontal circular pipe of diameter D,

$$\Delta P = \frac{32\mu L V_{avg}}{D^2}$$

The pressure drop for all type of flow, developed pipe flow (laminar or turbulent flows, circular or non-circular pipes, smooth or rough surfaces, horizontal

A. Pump speed necessarily halves

B. Pipe diameter doubles

C. The rising system head shifts each pump to lower individual discharge

D. Parallel pumps halve static head

Embedded figure file: images/diagrams/civil-flt02/q35-pump-characteristic.jpg

Civil FLT-02 · `data/civil/ce-flt02.js#Q41` · Key C

A tension plate has \(A_g=1800\) mm², \(A_n=1450\) mm², \(f_y=250\) MPa, \(f_u=410\) MPa; take \(\gamma_{m0}=1.10,\gamma_{m1}=1.25\). Compare gross yielding \(A_gf_y/\gamma_{m0}\) with net rupture \(0.9A_nf_u/\gamma_{m1}

97) The plane truss shown in Fig.2 carries a point load W and a moment M at the location B. Force carried by member AB is ________.

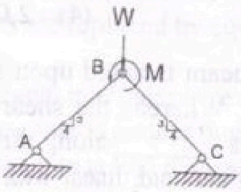


Fig.2

(1) 5W/6 (compressive) (2) W/2 (compressive)
(3) 5W/6 + M/L (compressive) (4) W/2 − M/2 (compressive)

16

A. yielding 450 kN governs

B. rupture 350 kN governs

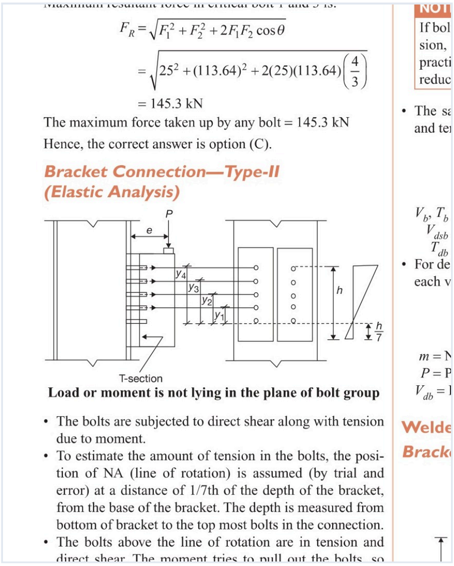
C. yielding 409 kN governs

D. rupture 428 kN governs

Embedded figure file: images/diagrams/civil-flt02/q41-plane-truss.jpg

Civil FLT-02 · `data/civil/ce-flt02.js#Q43` · Key D

The staggered plate crop shows two candidate net-section rupture paths P and Q. Using \(b=240\) mm, \(t=10\) mm, hole diameter 22 mm and the shown stagger/pitch values, which path and net area govern under \(b_n=b-\Sigma


$$F_R = \sqrt{F_1^2 + F_2^2 + 2F_1F_2\cos\theta}$$

$$= \sqrt{25^2 + (113.64)^2 + 2(25)(113.64)\left(\frac{4}{3}\right)}$$

$$= 145.3 \text{ kN}$$

The maximum force taken up by any bolt = 145.3 kN

Hence, the correct answer is option (C).

Bracket Connection—Type-II (Elastic Analysis)

Load or moment is not lying in the plane of bolt group

- The bolts are subjected to direct shear along with tension due to moment.
- To estimate the amount of tension in the bolts, the position of NA (line of rotation) is assumed (by trial and error) at a distance of 1/7th of the depth of the bracket, from the base of the bracket. The depth is measured from bottom of bracket to the top most bolts in the connection.
- The bolts above the line of rotation are in tension and

A. P, 1960 mm²

B. P, 1740 mm²

C. Q, 2030 mm²

D. Q, 1810 mm²

Embedded figure file: images/diagrams/civil-flt02/q43-eccentric-bolts.jpg

Elimination Rule 2 — Service vs factored / partial-factor mix

Evidence: ESTABLISHED — 104 independent (102 bank + 2 VALID PYQ) · **Streams:** Civil

What this rule means

Limit-state design mixes service loads, factored loads, and material strengths (fy, fck). A classic trap is to treat fy as the computed member stress, or to apply load factors twice.

When to use

Stem mixes service loads with limit-state options, or quotes fy/fck beside Mu/Vu.

Application steps

1. Label the stem: SLS (service) or ULS (factored Mu/Vu).
2. If Mu/Vu is already factored, do not multiply by gamma_f again.
3. Strike options that equal fy or 0.87 fy when the ask is a computed stress from P/A or M/Z.
4. Keep Ast calculations on the factored Mu and design stress 0.87 fy (or the code path stated).

```
Memory cue: ULS actions use gamma_f; design steel stress is typically 0.87 fy
(IS 456 path)
```

Core elimination move

Ask: is this SLS or ULS? Strike options that apply γf to an already-factored action, or that treat fy as the computed stress.

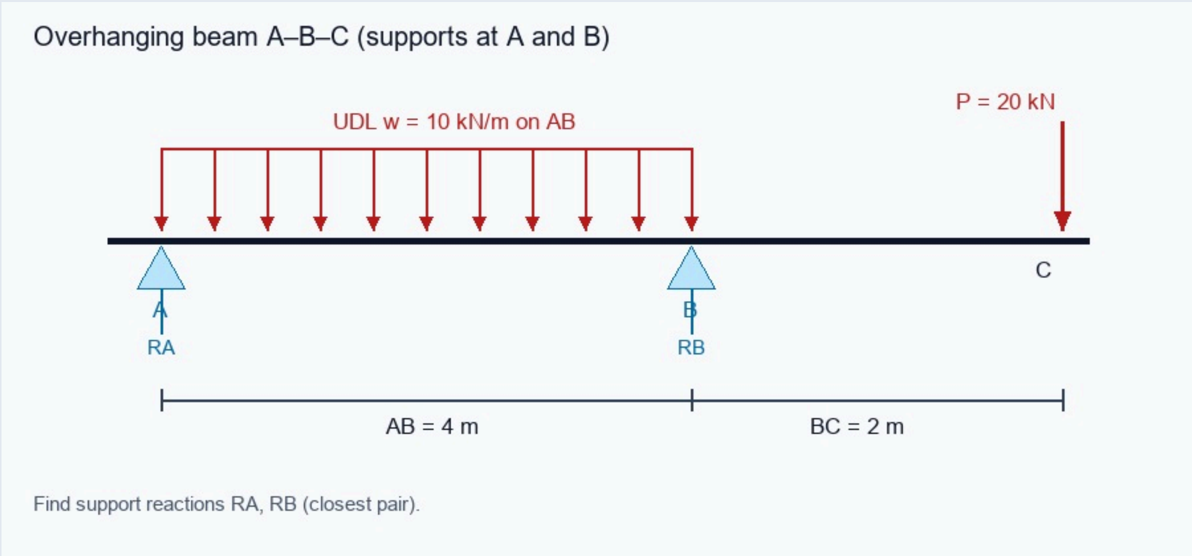
Evidence examples (sample references)

Source exam	Reference	Stem (truncated)
Civil FLT-01	`data/civil/ce-flt01.js#Q1`	Following a gantry-tie inspection at a substation, the engineer records a 20 mm diameter member under 55 kN…
Civil FLT-01	`data/civil/ce-flt01.js#Q3`	Proof-load planning uses an overhanging beam A–B–C with supports at A and B: AB = 4 m, overhang BC = 2 m, U…
Civil FLT-01	`data/civil/ce-flt01.js#Q7`	During review of a stepped shaft ABC, an end torque of 1000 N·m is applied at C while B is an intermediate …
Civil FLT-02	`data/civil/ce-flt02.js#Q1`	A 2.0 m long steel tie of uniform 500 mm^2 area carries a gradually applied 50 kN service load. Neglecting s…
Civil ST-FE	`data/civil/st/ce-st-fe-01.js#Q3`	The shown square footing distributes a concentric service load of 260 kN over a 2.3 m × 2.3 m base. The ave…
Civil ST-RCC	`data/civil/st/ce-st-rcc-01.js#Q1`	An RCC beam of M25 and Fe500 has factored Mu = 180 kN m, b = 230 mm and d = 450 mm. Using z = 0.9d, the req…

Figure evidence (actual diagrams — not path titles)

Civil FLT-01 · `data/civil/ce-flt01.js#Q3` · Key A

Proof-load planning uses an overhanging beam A–B–C with supports at A and B: AB = 4 m, overhang BC = 2 m, UDL w = 10 kN/m over AB, and a tip load P = 20 kN at C (also shown on the figure). Which support-reaction pair (RA



A. 10 kN, 50 kN

B. 15 kN, 45 kN

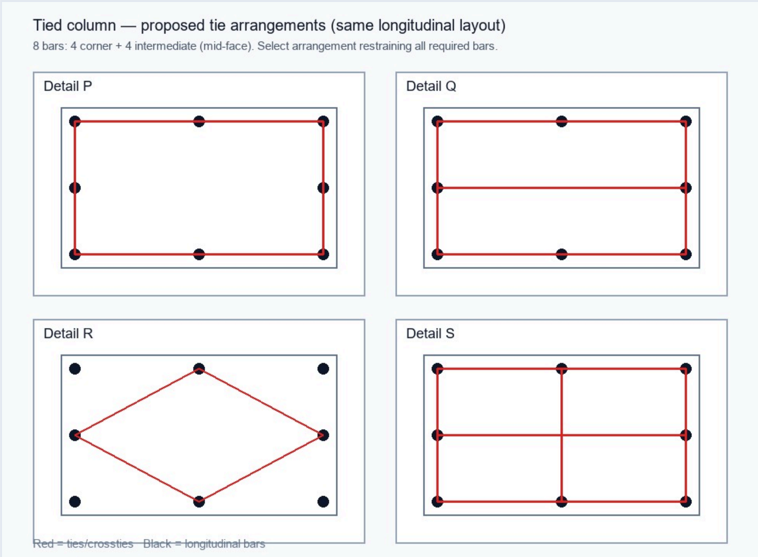
C. 20 kN, 40 kN

D. 25 kN, 35 kN

Embedded figure file: images/diagrams/civil-flt01/q03-overhang-reactions.jpg

Civil FLT-01 · `data/civil/ce-flt01.js#Q21` · Key D

At reinforcement inspection, a tied column shows the longitudinal-bar layout with intermediate bars on each face, and four proposed tie arrangements labelled Detail P, Q, R and S on the figure. Which marked arrangement b



A. Detail P

B. Detail Q

C. Detail R

D. Detail S

Embedded figure file: images/diagrams/civil-flt01/q21-column-tie-detail.jpg

Civil FLT-01 · `data/civil/ce-flt01.js#Q26` · Key D

From the design shear table (also shown), τc = 0.48 MPa at 0.50% tension steel and τc = 0.56 MPa at 0.75%. For pt = 0.625% and nominal shear stress τv = 0.70 MPa, which conclusion is closest?

he maximum moments per unit width are given by the
llowing equations:

$$\boxed{\begin{aligned} M_x &= \alpha_x w l_x^2 \\ M_y &= \alpha_y w l_x^2 \end{aligned}}$$

here

M_x and M_y are design moments along the short and long spans.

w = Uniformly distributed load on the slab.

l_x and l_y = The lengths of short and long spans.

α_x and α_y = The moment coefficients.

mply supported on four sides (Table 27 of IS 456–2000)

1.5	1.75	2.0	2.5	3.0
0.104	0.113	0.118	0.122	0.124
0.046	0.057	0.029	0.020	0.014

A. 0.48 MPa; minimum stirrups only

B. 0.56 MPa; no stirrups

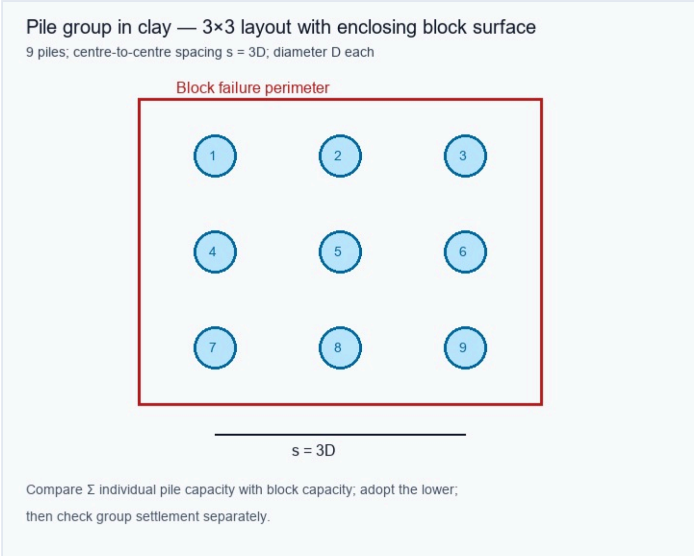
C. 0.70 MPa; concrete alone

D. 0.52 MPa; stirrups carry the balance, subject to the maximum limit

Embedded figure file: images/diagrams/civil-flt01/q26-shear-table.jpg

Civil FLT-01 · `data/civil/ce-flt01.js#Q57` · Key B

A pile group in clay uses a closely spaced 3×3 layout of 9 piles at centre-to-centre spacing s = 3D, with the enclosing block failure surface marked on the figure. Which capacity procedure is most defensible?



A. Use individual-pile sum only

B. Compare individual sum with block capacity, adopt the lower, then check settlement

C. Use block capacity only and ignore settlement

D. Assume group efficiency exceeds unity

Embedded figure file: images/diagrams/civil-flt01/q57-pile-group.jpg

Elimination Rule 3 — Gross vs net / hole deduction

Evidence: ESTABLISHED — 95 independent (95 bank + 0 VALID PYQ) · **Streams:** Civil

What this rule means

Tension capacity of bolted plates uses net area, not gross. For staggered holes, add s^2/(4g) terms. Options that equal gross area ignore the holes entirely.

When to use

Tension member, plate with bolt holes, staggered pitch, or chain of holes.

Application steps

1. Write Anet = (gross width - n*d_hole)*t (+ stagger terms if shown).
2. Strike the gross-area option immediately.
3. If stagger is drawn, include s^2/(4g); if chain is transverse only, do not invent stagger credit.

```
Memory cue: Anet ~ (b - n*dh)*t ; staggered: + sum s^2/(4g)
```

Core elimination move

Net = gross − holes (± $s^2/4g$). Eliminate options that equal gross area, or that deduct diameter without adding the staggered term when stagger is shown.

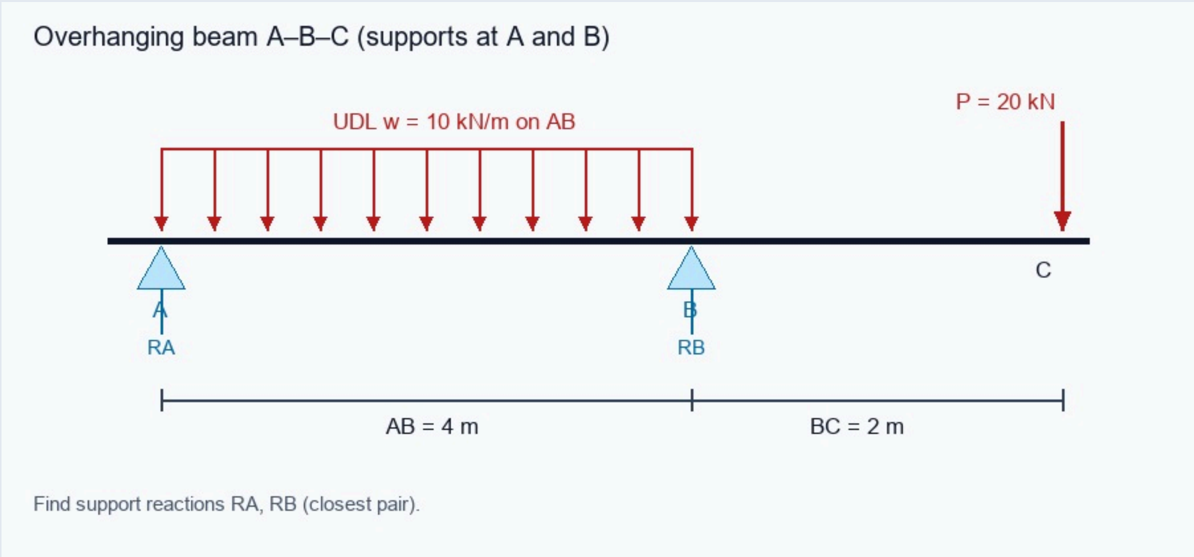
Evidence examples (sample references)

Source exam	Reference	Stem (truncated)
Civil FLT-01	`data/civil/ce-flt01.js#Q1`	Following a gantry-tie inspection at a substation, the engineer records a 20 mm diameter member under 55 kN...
Civil FLT-01	`data/civil/ce-flt01.js#Q3`	Proof-load planning uses an overhanging beam A–B–C with supports at A and B: AB = 4 m, overhang BC = 2 m, U...
Civil FLT-01	`data/civil/ce-flt01.js#Q7`	During review of a stepped shaft ABC, an end torque of 1000 N·m is applied at C while B is an intermediate ...
Civil FLT-02	`data/civil/ce-flt02.js#Q20`	Two short tied columns have the same gross area and longitudinal-steel percentage. Column X alone receives ...
Civil ST-FE	`data/civil/st/ce-st-fe-01.js#Q1`	For a strip footing, take c = 20 kPa, Nc = 25, q = 30 kPa, Nq = 12, γ = 18 kN/m³, B = 2 m and Nγ = 8. Terza...
Civil ST-RCC	`data/civil/st/ce-st-rcc-01.js#Q4`	A 300 mm × 300 mm short tied column carries factored axial load 1050 kN. The average factored stress Pu/Ag ...

Figure evidence (actual diagrams — not path titles)

Civil FLT-01 · `data/civil/ce-flt01.js#Q3` · Key A

Proof-load planning uses an overhanging beam A–B–C with supports at A and B: AB = 4 m, overhang BC = 2 m, UDL w = 10 kN/m over AB, and a tip load P = 20 kN at C (also shown on the figure). Which support-reaction pair (RA



A. 10 kN, 50 kN

B. 15 kN, 45 kN

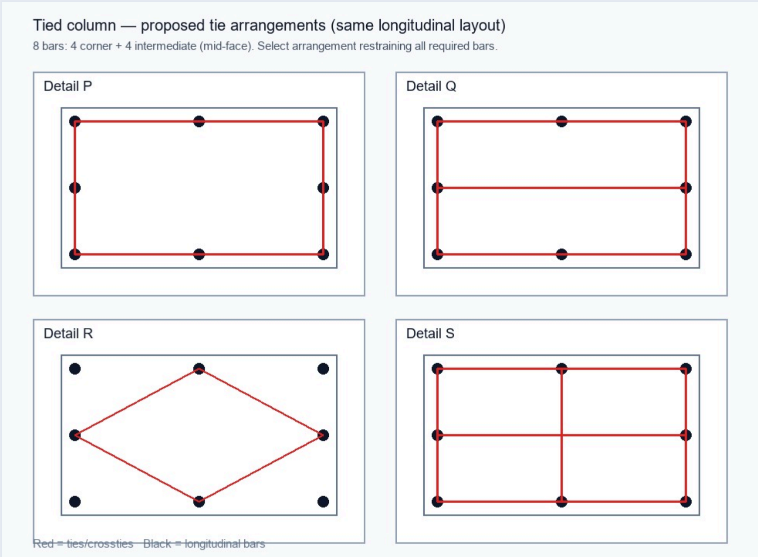
C. 20 kN, 40 kN

D. 25 kN, 35 kN

Embedded figure file: images/diagrams/civil-flt01/q03-overhang-reactions.jpg

Civil FLT-01 · `data/civil/ce-flt01.js#Q21` · Key D

At reinforcement inspection, a tied column shows the longitudinal-bar layout with intermediate bars on each face, and four proposed tie arrangements labelled Detail P, Q, R and S on the figure. Which marked arrangement b



A. Detail P

B. Detail Q

C. Detail R

D. Detail S

Embedded figure file: images/diagrams/civil-flt01/q21-column-tie-detail.jpg

Civil FLT-01 · `data/civil/ce-flt01.js#Q26` · Key D

From the design shear table (also shown), τc = 0.48 MPa at 0.50% tension steel and τc = 0.56 MPa at 0.75%. For pt = 0.625% and nominal shear stress τv = 0.70 MPa, which conclusion is closest?

he maximum moments per unit width are given by the
llowing equations:

$$\boxed{\begin{aligned} M_x &= \alpha_x w l_x^2 \\ M_y &= \alpha_y w l_x^2 \end{aligned}}$$

'here

M_x and M_y are design moments along the short and long spans.

w = Uniformly distributed load on the slab.

l_x and l_y = The lengths of short and long spans.

α_x and α_y = The moment coefficients.

mply supported on four sides (Table 27 of IS 456–2000)

1.5	1.75	2.0	2.5	3.0
0.104	0.113	0.118	0.122	0.124
0.046	0.057	0.029	0.020	0.014

A. 0.48 MPa; minimum stirrups only

B. 0.56 MPa; no stirrups

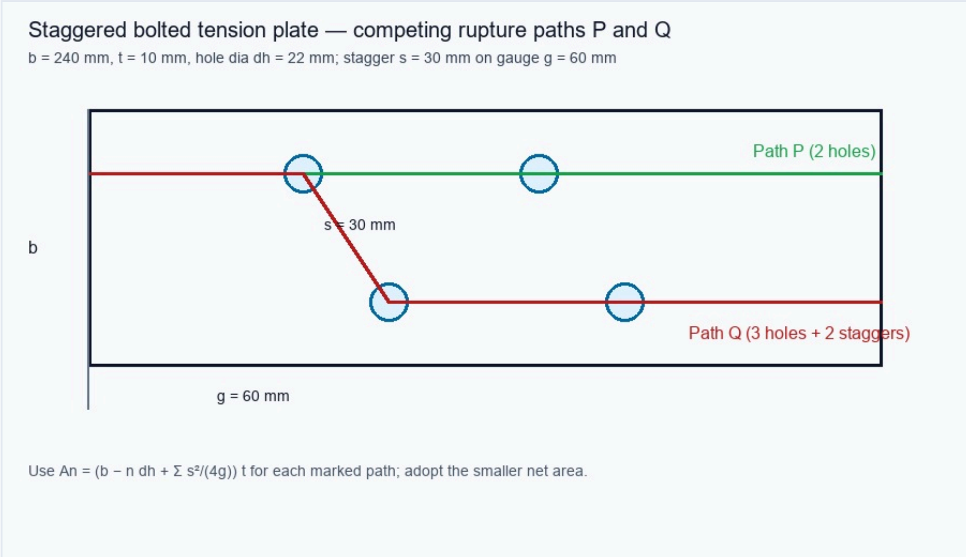
C. 0.70 MPa; concrete alone

D. 0.52 MPa; stirrups carry the balance, subject to the maximum limit

Embedded figure file: images/diagrams/civil-flt01/q26-shear-table.jpg

Civil FLT-01 · `data/civil/ce-flt01.js#Q47` · Key D

A bolted tension plate has width b = 240 mm, thickness t = 10 mm and hole diameter dh = 22 mm. Path P is a chain section through 2 holes; path Q zig-zags through 3 holes with two stagger credits s = 30 mm on gauge g = 60



A. P, 1960 mm^2

B. P, 1740 mm^2

C. Q, 2030 mm^2

D. Q, 1810 mm^2

Embedded figure file: images/diagrams/civil-flt01/q47-staggered-net-paths.jpg

Elimination Rule 4 — Cover vs effective depth

Evidence: ESTABLISHED — 40 independent (39 bank + 1 VALID PYQ) · **Streams:** Civil

What this rule means

RCC flexural and shear formulas use effective depth d, not overall depth D. Cover and bar diameter set d = D - cover - phi/2 (as defined in the stem).

When to use

RCC beam/slab with overall D, cover, bar dia — options in Ast, τv, or lever arm.

Application steps

1. Extract D, cover, and bar size; compute d before Ast or tau_v.
2. Strike options that use D in place of d in Mu = 0.87 fy Ast z or tau_v = Vu/(b d).

```
Memory cue: d = D - cover - phi/2 ; tau_v = Vu/(b d)
```

Core elimination move

d = D − cover − φ/2 (or as stem defines). Strike options that use overall depth as effective depth.

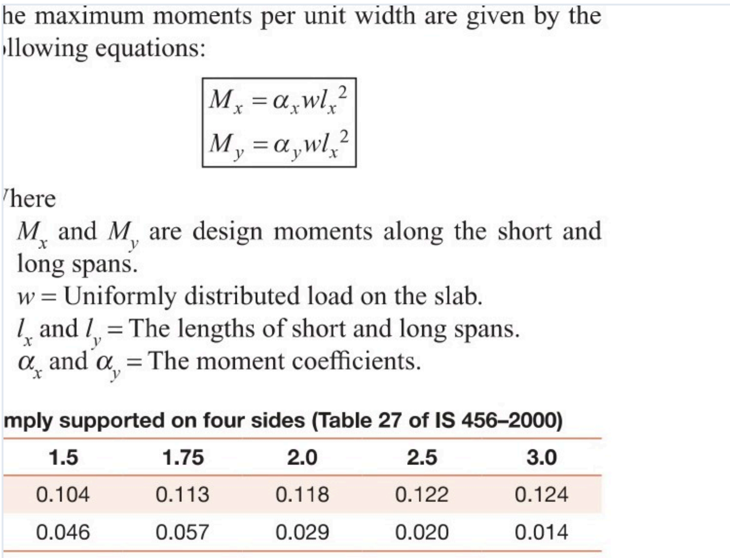
Evidence examples (sample references)

Source exam	Reference	Stem (truncated)
Civil FLT-01	`data/civil/ce-flt01.js#Q16`	Before a pour on an RCC beam in moderate exposure, measured cover is below the durability requirement altho...
Civil FLT-01	`data/civil/ce-flt01.js#Q17`	While checking an Fe415 singly reinforced beam, the trial neutral-axis depth is 225 mm for an effective dep...
Civil FLT-01	`data/civil/ce-flt01.js#Q19`	Mid-pour on a continuous RCC floor, top bars over an interior support are found displaced downward by about...
Civil FLT-02	`data/civil/ce-flt02.js#Q16`	An RCC beam in moderate exposure is detailed with nominal cover smaller than the durability requirement, th...
Civil ST-RCC	`data/civil/st/ce-st-rcc-01.js#Q1`	An RCC beam of M25 and Fe500 has factored Mu = 180 kN m, b = 230 mm and d = 450 mm. Using z = 0.9d, the req...
Civil ST-STEEL	`data/civil/st/ce-st-steel-01.js#Q4`	A lap connection has 4 bearing-type bolts, each with governing design shear capacity 48 kN. Ignoring eccent...

Figure evidence (actual diagrams — not path titles)

Civil FLT-01 · `data/civil/ce-flt01.js#Q26` · Key D

From the design shear table (also shown), τc = 0.48 MPa at 0.50% tension steel and τc = 0.56 MPa at 0.75%. For pt = 0.625% and nominal shear stress τv = 0.70 MPa, which conclusion is closest?


he maximum moments per unit width are given by the
ıllowing equations:

$$\boxed{\begin{aligned} M_x &= \alpha_x w l_x^2 \\ M_y &= \alpha_y w l_x^2 \end{aligned}}$$

'here
M_x and M_y are design moments along the short and long spans.
w = Uniformly distributed load on the slab.
l_x and l_y = The lengths of short and long spans.
α_x and α_y = The moment coefficients.

mply supported on four sides (Table 27 of IS 456–2000)

1.5	1.75	2.0	2.5	3.0
0.104	0.113	0.118	0.122	0.124
0.046	0.057	0.029	0.020	0.014

A. 0.48 MPa; minimum stirrups only

B. 0.56 MPa; no stirrups

C. 0.70 MPa; concrete alone

D. 0.52 MPa; stirrups carry the balance, subject to the maximum limit

Embedded figure file: images/diagrams/civil-flt01/q26-shear-table.jpg

Civil FLT-02 · `data/civil/ce-flt02.js#Q23` · Key D

The crop shows a continuous-beam bar proposed to terminate at section P, with the theoretical cut-off, support face and remaining anchorage marked only in the figure. Which option correctly decides acceptance after check

A) Rs. 750
B) Rs. 1000
C) Rs. 800
D) Rs. 1200
Ans: **B (revised)**

100. In adverse circumstances, the reinforced concrete member immersed in sea water or subjected to sea spray, the maximum permissible cover for the reinforcing bars, should not exceed
A) 40 mm
B) 50 mm
C) 60 mm
D) 75mm
Ans: D

A. Accept if cover is adequate

B. Reject every curtailment

C. Accept solely because P is beyond inflection

D. Accept only if all three checks and extension rules pass

Embedded figure file: images/diagrams/civil-flt02/q23-bar-curtailment.jpg

Elimination Rule 5 — Per-unit base-change scaling

Evidence: ESTABLISHED — 95 independent (95 bank + 0 VALID PYQ) · **Streams:** Electrical

What this rule means

Per-unit impedance scales with the chosen bases: Zpu = Zohm * Sbase / Vbase^2. Changing only Sbase scales Zpu in direct proportion.

When to use

Zpu / Xpu and base MVA or kV changes.

Application steps

1. Note old and new Sbase and Vbase.
2. If Sbase doubles and Vbase is unchanged, Zpu halves — strike 'unchanged', 'doubles', 'squared'.
3. If Vbase also changes, recompute the full ratio.

```
Memory cue: Zpu = Zohm * Sbase / Vbase^2
```

Core elimination move

Zpu ∝ Sbase / Vbase². If Sbase doubles and Vbase fixed → Zpu halves. Eliminate 'unchanged', 'doubles', or 'squared' unless the stem changed Vbase too.

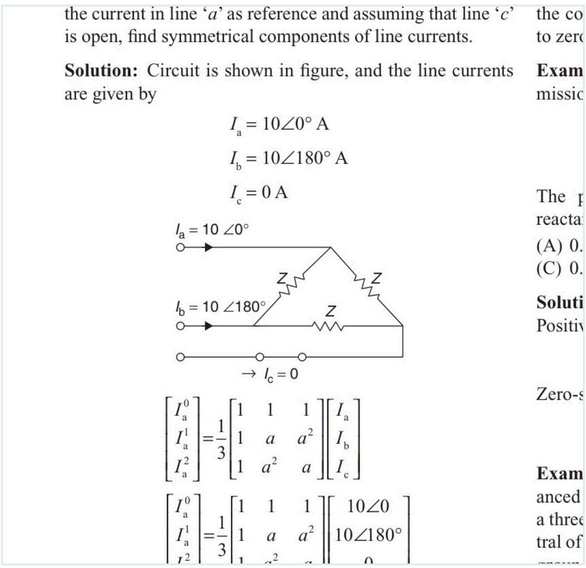
Evidence examples (sample references)

Source exam	Reference	Stem (truncated)
Electrical FLT-01	`data/electrical/ee-flt01.js#Q1`	A lossless 400 kV line has Zc = 250 Ω. Its SIL (MW) is closest to:
Electrical FLT-01	`data/electrical/ee-flt01.js#Q2`	On 100 MVA, 220 kV base, a line reactance 48.4 Ω equals how many pu?
Electrical FLT-01	`data/electrical/ee-flt01.js#Q3`	A 3-φ feeder has positive-sequence impedance j0.2 pu and zero-sequence j0.5 pu on same base. For a solid SL...
Electrical FLT-02	`data/electrical/ee-flt02.js#Q2`	On 200 MVA, 400 kV base, a series reactor is 96 ohms. Per-unit reactance is closest to:
Electrical ST-PS	`data/electrical/st/ee-st-ps-01.js#Q1`	On a common 100 MVA base, if the MVA base is doubled while the base kV is unchanged, a line reactance expre...

Figure evidence (actual diagrams — not path titles)

Electrical FLT-01 · `data/electrical/ee-flt01.js#Q9` · Key A

From the open-conductor / sequence figure shown (use only labelled phase currents; Ic = 0), the zero-sequence current Ia0 is:



A. 0 A

B. 5.78 A

C. 3.33 A

D. 10 A

Embedded figure file: images/diagrams/electrical-flt01/q09-open-conductor.jpg

Electrical FLT-01 · `data/electrical/ee-flt01.js#Q18` · Key B

From the alternator phasor diagram shown (Ia leading Vt), the operating power factor is:

ator

$$\text{KVA} = \frac{\sqrt{3}\, V_{LL} I_{fl}}{1000}$$

$$\Rightarrow 1000 \times 1000 = \sqrt{3} \times 2000 \times i_{fl}$$

$$\Rightarrow I_{fc} = 288.7 = I_a$$

at 0.8 leading

$$E = V_p + I_a R_a = 1155 + (288.7 \times 0.2 \angle \cos^{-1}(0.8))$$

$$= 1155 + 46.2 + j34.44 = 1201.7 \angle 1.6°$$

$$I_{FI} = I_F \angle(90 + \alpha) = 32 \angle(90 + 1.6°)$$

(Ampere-turns method)

$$= -0.89 + j\,31.98 \text{ A}$$

$$I_{f2} = 29 \angle(180 + 36.87°) = 29\angle 216.87° \text{A}$$

$$I_f = I_{f1} + I_{f2} = 28.15 \angle 131.18°$$

at this I_f, $E = 1098$ V

$$\Rightarrow V_R = \frac{10998 - 1155}{1155}$$

$$= -5.02\%$$

Example 4: A 3-phase alternator delivers power to a

A. Unity only

B. Leading pf load

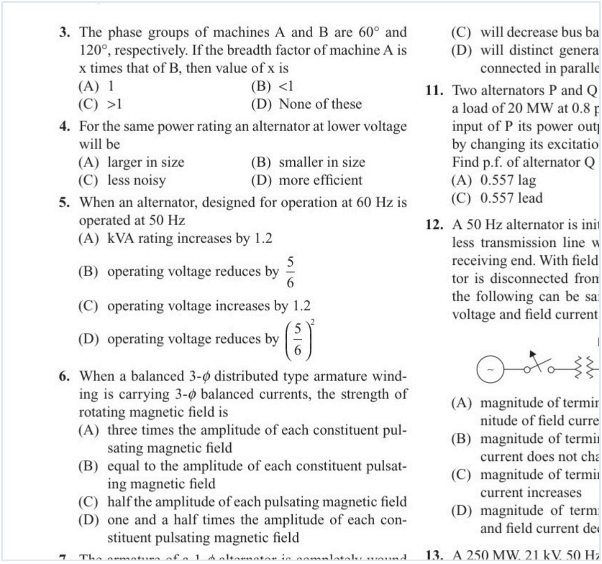
C. Short-circuit only

D. Lagging pf load

Embedded figure file: images/diagrams/electrical-flt01/q18-leading-phasor.jpg

Electrical FLT-01 · `data/electrical/ee-flt01.js#Q22` · Key B

A 50 Hz alternator is connected to a long lossless line open at the receiving end as shown. With field voltage held constant, the generator is disconnected from the line. Steady |Vt|:



3. The phase groups of machines A and B are 60° and 120°, respectively. If the breadth factor of machine A is x times that of B, then value of x is
(A) 1 (B) <1
(C) >1 (D) None of these

4. For the same power rating an alternator at lower voltage will be
(A) larger in size (B) smaller in size
(C) less noisy (D) more efficient

5. When an alternator, designed for operation at 60 Hz is operated at 50 Hz
(A) kVA rating increases by 1.2
(B) operating voltage reduces by $\frac{5}{6}$
(C) operating voltage increases by 1.2
(D) operating voltage reduces by $\left(\frac{5}{6}\right)^2$

6. When a balanced 3-ϕ distributed type armature winding is carrying 3-ϕ balanced currents, the strength of rotating magnetic field is
(A) three times the amplitude of each constituent pulsating magnetic field
(B) equal to the amplitude of each constituent pulsating magnetic field
(C) half the amplitude of each pulsating magnetic field
(D) one and a half times the amplitude of each constituent pulsating magnetic field

(C) will decrease bus ba
(D) will distinct genera
connected in paralle

11. Two alternators P and Q
a load of 20 MW at 0.8 p
input of P its power out
by changing its excitatio
Find p.f. of alternator Q
(A) 0.557 lag
(C) 0.557 lead

12. A 50 Hz alternator is ini
less transmission line w
receiving end. With field
tor is disconnected from
the following can be sa
voltage and field current

(A) magnitude of termin
nitude of field curre
(B) magnitude of termi
current does not cha
(C) magnitude of termi
current increases
(D) magnitude of term
and field current de

13. A 250 MW, 21 kV, 50 Hz

A. Unchanged always

B. Decreases (Ferranti charging removed)

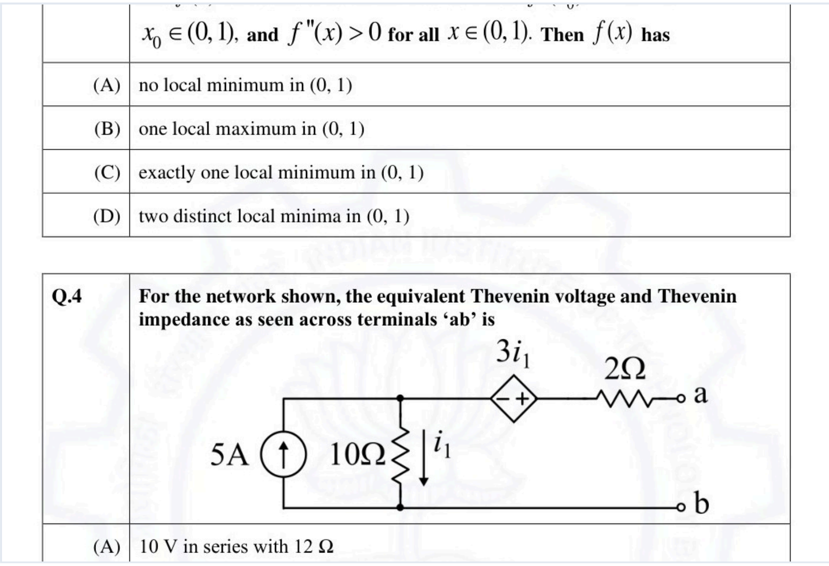
C. Becomes zero always

D. Increases always

Embedded figure file: images/diagrams/electrical-flt01/q22-ferranti-line.jpg

Electrical FLT-01 · `data/electrical/ee-flt01.js#Q33` · Key A

From the frequency-response plot shown (pass band in the middle with stop bands on both sides), the filter type is:



	$x_0 \in (0,1)$, **and** $f''(x) > 0$ **for all** $x \in (0,1)$. **Then** $f(x)$ **has**
(A)	no local minimum in (0, 1)
(B)	one local maximum in (0, 1)
(C)	exactly one local minimum in (0, 1)
(D)	two distinct local minima in (0, 1)

Q.4	**For the network shown, the equivalent Thevenin voltage and Thevenin impedance as seen across terminals 'ab' is**
(A)	10 V in series with 12 Ω

A. Band-pass

B. High-pass

C. Band-elimination

D. Low-pass

Embedded figure file: images/diagrams/electrical-flt01/q33-filter-plot.jpg

Elimination Rule 6 — Sequence-network connection by fault type

Evidence: ESTABLISHED — 25 independent (18 bank + 7 VALID PYQ) · **Streams:** Electrical

What this rule means

Unsymmetrical faults require specific sequence-network connections. Memorising the four classic connections eliminates most wrong options instantly.

When to use

LG / LL / LLG / 3φ fault; options name sequence networks.

Application steps

1. Identify fault type: 3-ph, LG, LL, LLG.
2. 3-ph → positive only; LG → 1-2-0 in series; LL → 1 parallel 2; LLG → 1 series (2 parallel 0).
3. Strike any option that omits a required sequence or uses only zero-sequence.

```
Memory cue: LG: Z1+Z2+Z0 ; LL: Z1||Z2 ; LLG: Z1+(Z2||Z0)
```

Core elimination move

3φ → positive only. LG → series 1-2-0. LL → 1 parallel 2. LLG → 1 series (2‖0). Eliminate any option that omits a required sequence or uses only zero-sequence.

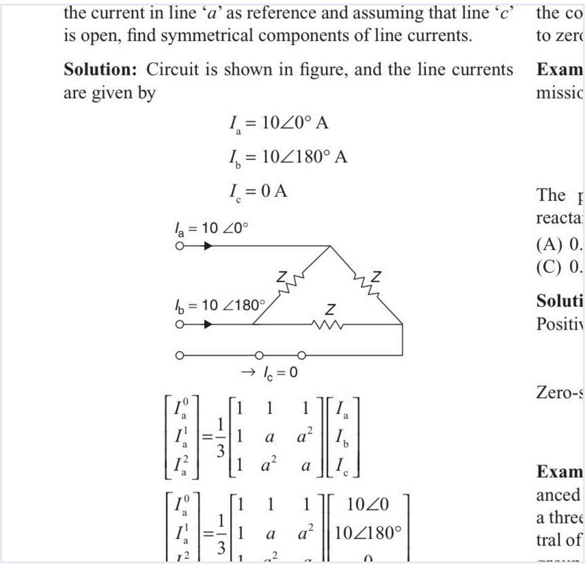
Evidence examples (sample references)

Source exam	Reference	Stem (truncated)
Electrical FLT-01	`data/electrical/ee-flt01.js#Q3`	A 3-φ feeder has positive-sequence impedance j0.2 pu and zero-sequence j0.5 pu on same base. For a solid SL...
Electrical FLT-01	`data/electrical/ee-flt01.js#Q8`	For a symmetrical 3-φ fault, sequence networks are connected in:
Electrical FLT-01	`data/electrical/ee-flt01.js#Q9`	From the open-conductor / sequence figure shown (use only labelled phase currents; Ic = 0), the zero-sequen...
Electrical FLT-02	`data/electrical/ee-flt02.js#Q3`	Solid LG fault: Z1=Z2=j0.15 pu, Z0=j0.60 pu, Vpref=1 pu. |If| (pu) nearest:
Electrical ST-EM	`data/electrical/st/ee-st-em-01.js#Q17`	A synchronous generator is being synchronised with live busbars. Which statement is correct?
Electrical ST-PS	`data/electrical/st/ee-st-ps-01.js#Q2`	For a double line-to-ground fault on an unloaded generator, the sequence networks required for the fault ca...

Figure evidence (actual diagrams — not path titles)

Electrical FLT-01 · `data/electrical/ee-flt01.js#Q9` · Key A

From the open-conductor / sequence figure shown (use only labelled phase currents; Ic = 0), the zero-sequence current Ia0 is:


the current in line '*a*' as reference and assuming that line '*c*' is open, find symmetrical components of line currents.

Solution: Circuit is shown in figure, and the line currents are given by

$$I_a = 10\angle 0^\circ \text{ A}$$
$$I_b = 10\angle 180^\circ \text{ A}$$
$$I_c = 0 \text{ A}$$

$$\begin{bmatrix} I_a^0 \\ I_a^1 \\ I_a^2 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix}$$

$$\begin{bmatrix} I_a^0 \\ I_a^1 \\ I_a^2 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}\begin{bmatrix} 10\angle 0 \\ 10\angle 180^\circ \\ 0 \end{bmatrix}$$

A. 0 A

B. 5.78 A

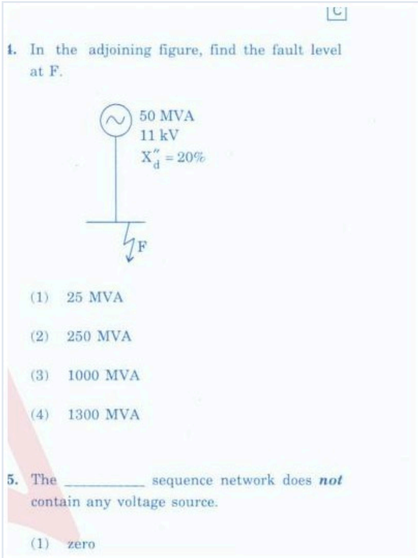
C. 3.33 A

D. 10 A

Embedded figure file: images/diagrams/electrical-flt01/q09-open-conductor.jpg

Electrical FLT-02 · `data/electrical/ee-flt02.js#Q9` · Key B

From the sequence-network sketch (open conductor on phase C, Ia=Ib=5 A, Ic=0), the zero-sequence component Ia0 equals:



4. In the adjoining figure, find the fault level at F.

50 MVA
11 kV
$X''_d = 20\%$

F

(1) 25 MVA

(2) 250 MVA

(3) 1000 MVA

(4) 1300 MVA

5. The ________ sequence network does *not* contain any voltage source.

(1) zero

A. 5 A

B. 0 A

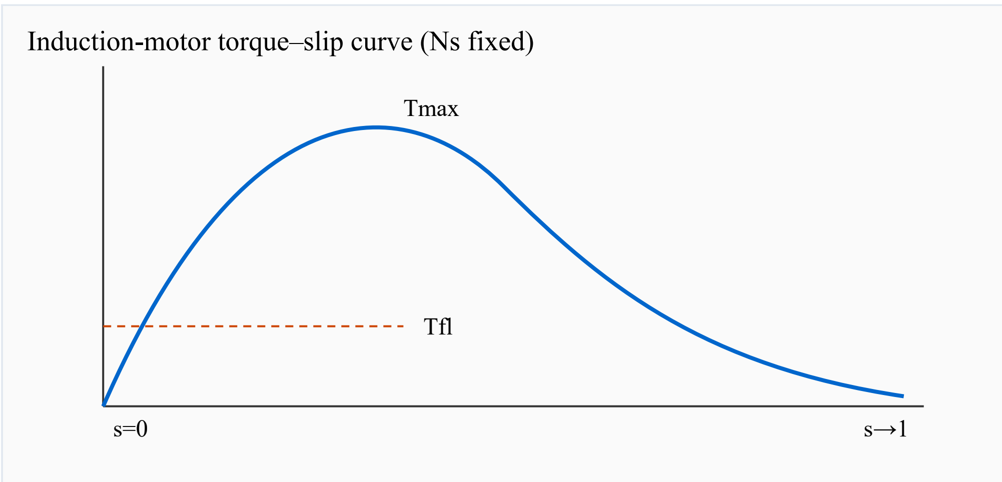
C. 10 A

D. 3.33 A

Embedded figure file: images/diagrams/electrical-flt02/q09-sequence-network.jpg

Electrical ST-EM · `data/electrical/st/ee-st-em-01.js#Q47` · Key D

A synchronous generator is being synchronised with live busbars. Which statement is correct?



A. Only voltage magnitude needs to match.

B. Frequency may differ if kVA ratings match.

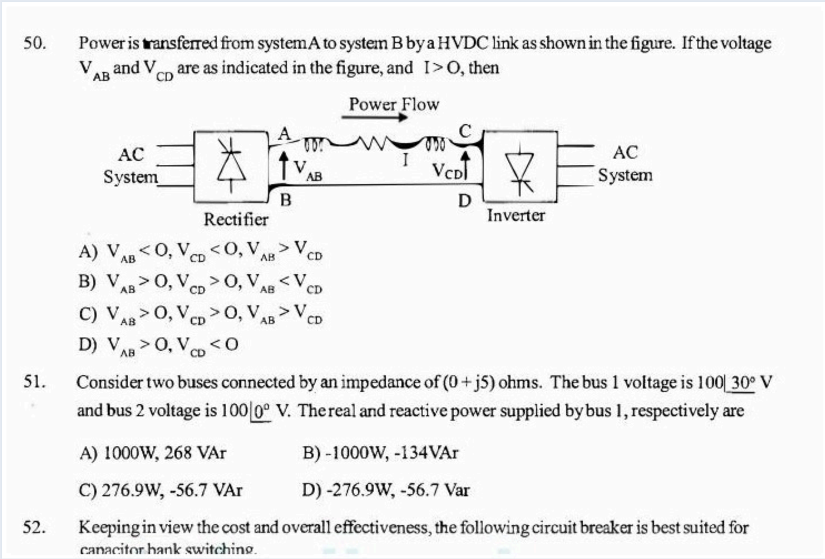
C. Phase sequence is immaterial.

D. Voltage magnitude, frequency, phase sequence, and phase angle must match.

Embedded figure file: images/diagrams/electrical-st-em/q07-im-torque-slip.svg

Electrical ST-PS · `data/electrical/st/ee-st-ps-01.js#Q2` · Key D

For a double line-to-ground fault on an unloaded generator, the sequence networks required for the fault calculation must be connected as:



50. Power is transferred from system A to system B by a HVDC link as shown in the figure. If the voltage V_{AB} and V_{CD} are as indicated in the figure, and $I > 0$, then

A) $V_{AB} < 0, V_{CD} < 0, V_{AB} > V_{CD}$
B) $V_{AB} > 0, V_{CD} > 0, V_{AB} < V_{CD}$
C) $V_{AB} > 0, V_{CD} > 0, V_{AB} > V_{CD}$
D) $V_{AB} > 0, V_{CD} < 0$

51. Consider two buses connected by an impedance of $(0 + j5)$ ohms. The bus 1 voltage is $100\angle 30°$ V and bus 2 voltage is $100\angle 0°$ V. The real and reactive power supplied by bus 1, respectively are

A) 1000W, 268 VAr B) -1000W, -134VAr
C) 276.9W, -56.7 VAr D) -276.9W, -56.7 Var

52. Keeping in view the cost and overall effectiveness, the following circuit breaker is best suited for capacitor bank switching.

A. dc only

B. zero only

C. positive only

D. positive+negative+zero sequence networks

Embedded figure file: images/diagrams/electrical-st-ps/ps4.jpg

Elimination Rule 7 — √3 / phase-factor forgotten

Evidence: ESTABLISHED — 24 independent (18 bank + 6 VALID PYQ) · **Streams:** Electrical

What this rule means

Three-phase power and current conversions carry a root-3 factor. Forgetting it, or using 3 instead of root-3, produces the most common distractors.

When to use

3-phase power, line vs phase quantities, line current from kVA.

Application steps

1. Write P = root3 * VL * IL * cos(phi) (or the form asked).
2. Strike options that drop root3 or replace it with 3 without justification.

```
Memory cue: P = √3 · VL · IL · cosϕ
```

Core elimination move

P = √3 VL IL cosφ. Eliminate options that drop √3 or use 3 instead of √3 (or vice versa) without stem justification.

Evidence examples (sample references)

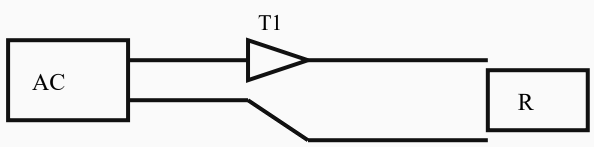
Source exam	Reference	Stem (truncated)
Electrical FLT-01	`data/electrical/ee-flt01.js#Q3`	A 3-φ feeder has positive-sequence impedance j0.2 pu and zero-sequence j0.5 pu on same base. For a solid SL…
Electrical FLT-01	`data/electrical/ee-flt01.js#Q5`	Two identical 50 MVA, 11 kV, Xd″=0.15 pu machines feed a bus. On 100 MVA base, the bus short-circuit MVA fo…
Electrical FLT-01	`data/electrical/ee-flt01.js#Q8`	For a symmetrical 3-φ fault, sequence networks are connected in:
Electrical FLT-02	`data/electrical/ee-flt02.js#Q8`	During a bolted three-phase short circuit on an unloaded radial feeder, which sequence network combination …
Electrical ST-EC	`data/electrical/st/ee-st-ec-01.js#Q17`	A balanced three-phase load is measured with the two-wattmeter method. Which statement is correct?
Electrical ST-MEAS	`data/electrical/st/ee-st-meas-01.js#Q8`	Two-wattmeter method pf=:

Figure evidence (actual diagrams — not path titles)

Electrical FLT-02 · `data/electrical/ee-flt02.js#Q48` · Key A

From the single-phase semiconverter schematic, average output voltage with firing angle α (0<α<180°) is proportional to:

Single-phase semi-converter, R-load, firing angle α



Vo,avg = (Vm/π)(1+cos α) for continuous conduction on R load

A. (1+cosα)

B. cos2α

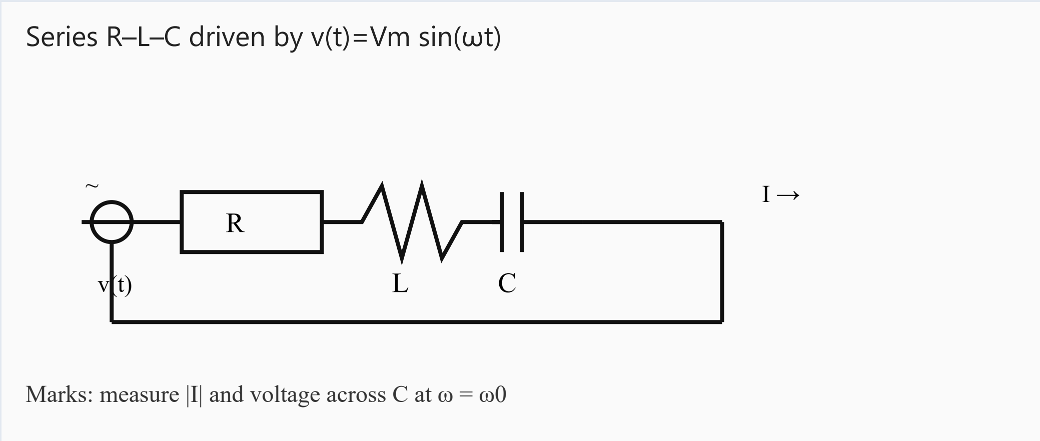
C. α in radians

D. sinα only

Embedded figure file: images/diagrams/electrical-flt02/q48-semi-converter.svg

Electrical ST-EC · `data/electrical/st/ee-st-ec-01.js#Q47` · Key A

A balanced three-phase load is measured with the two-wattmeter method. Which statement is correct?



A. The two readings sum to total active power.

B. Both wattmeters always read equal.

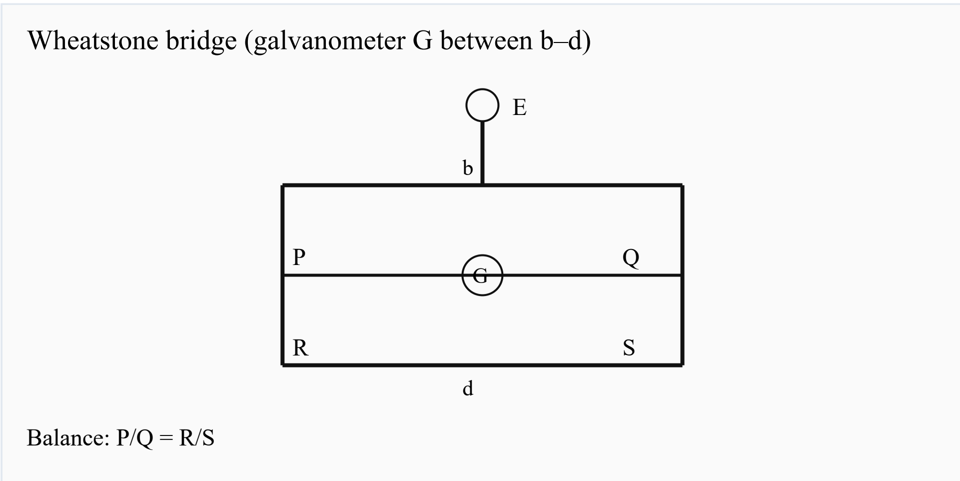
C. Their difference gives reactive power without a factor.

D. The method is unusable for balanced loads.

Embedded figure file: images/diagrams/electrical-st-ec/q07-series-rlc.svg

Electrical ST-MEAS · `data/electrical/st/ee-st-meas-01.js#Q8` · Key D

Two-wattmeter method pf=:



A. 1 always

B. 0

C. W1/W2

D. cos(tan^{-1}√3(W1−W2)/(W1+W2))

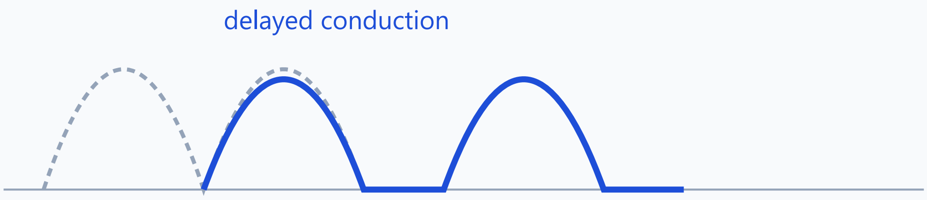
Embedded figure file: images/diagrams/electrical-st-meas/q07-wheatstone.svg

Electrical ST-PE · `data/electrical/st/ee-st-pe-01.js#Q9` · Key C

180° mode three-phase VSI line voltage steps:

Q48 — Semi-converter output waveform

1-φ semi-converter · α = 30° · RL load — Vo waveform



Vo stays non-negative with firing delay each half-cycle

A. pure sine without filter always

B. triangle only

C. six-step waveform

D. dc only

Embedded figure file: images/diagrams/electrical-st-pe/pe3.svg

Elimination Rule 8 — Thevenin / Norton source deactivation

Evidence: ESTABLISHED — 18 independent (14 bank + 4 VALID PYQ) · **Streams:** Electrical

What this rule means

Thevenin/Norton resistance is found with independent sources deactivated: voltage sources shorted, current sources opened.

When to use

Req, RN, or Vth with independent sources in the figure.

Application steps

1. For Req/RN, deactivate independent sources first.
2. Strike options that leave sources 'on' while claiming equivalent resistance.
3. Do not confuse Vth (open-circuit voltage) with RN.

```
Memory cue: Independent V → short; independent I → open; then find Req
```

Core elimination move

Independent voltage sources → short; current sources → open. Eliminate options that leave sources active while claiming Req/RN.

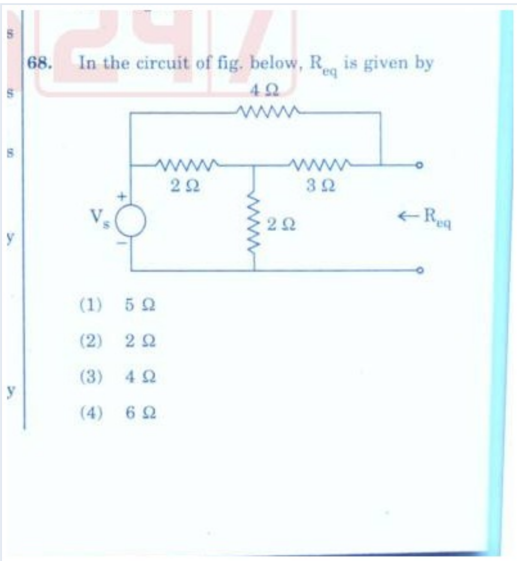
Evidence examples (sample references)

Source exam	Reference	Stem (truncated)
Electrical FLT-01	`data/electrical/ee-flt01.js#Q29`	In the resistor network shown driven by Vs, the equivalent resistance Req looking into the right-hand termi…
Electrical FLT-01	`data/electrical/ee-flt01.js#Q34`	An ideal ammeter is connected between terminals A and B in the circuit shown. The ammeter reading is:
Electrical FLT-01	`data/electrical/ee-flt01.js#Q56`	For the network shown, Norton's resistance RN seen at the open terminals is:
Electrical FLT-02	`data/electrical/ee-flt02.js#Q35`	Superposition theorem applies to:
Electrical ST-EC	`data/electrical/st/ee-st-ec-01.js#Q3`	After independent sources are deactivated, the Norton resistance seen at the output terminals is:
VALID PYQ · 2021EE.pdf	`PYQ:2021EE.pdf#Q4`	For the network shown, the equivalent Thevenin voltage and Thevenin impedance as seen across terminals 'ab' is

Figure evidence (actual diagrams — not path titles)

Electrical FLT-01 · `data/electrical/ee-flt01.js#Q29` · Key C

In the resistor network shown driven by Vs, the equivalent resistance Req looking into the right-hand terminals is:



A. 5 Ω

B. 2 Ω

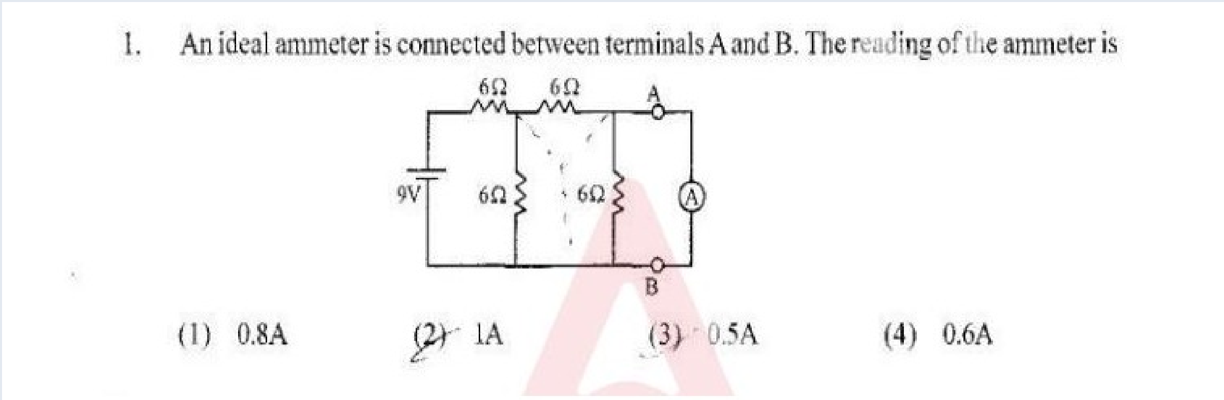
C. 4 Ω

D. 6 Ω

Embedded figure file: images/diagrams/electrical-flt01/q-extra-req.jpg

Electrical FLT-01 · `data/electrical/ee-flt01.js#Q34` · Key B

An ideal ammeter is connected between terminals A and B in the circuit shown. The ammeter reading is:



A. 0.8 A

B. 1 A

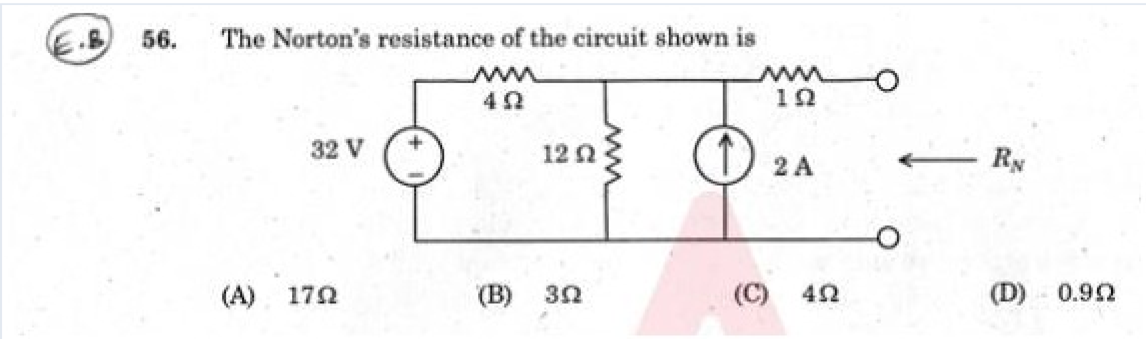
C. 0.5 A

D. 0.6 A

Embedded figure file: images/diagrams/electrical-flt01/q-extra-ammeter.jpg

Electrical FLT-01 · `data/electrical/ee-flt01.js#Q56` · Key B

For the network shown, Norton’s resistance RN seen at the open terminals is:



A. 17 Ω

B. 3 Ω

C. 4 Ω

D. 0.9 Ω

Embedded figure file: images/diagrams/electrical-flt01/q56-norton-circuit.jpg

Electrical FLT-02 · `data/electrical/ee-flt02.js#Q56` · Key D

From the Norton equivalent network diagram, after deactivating independent sources, RN across terminals is nearest:

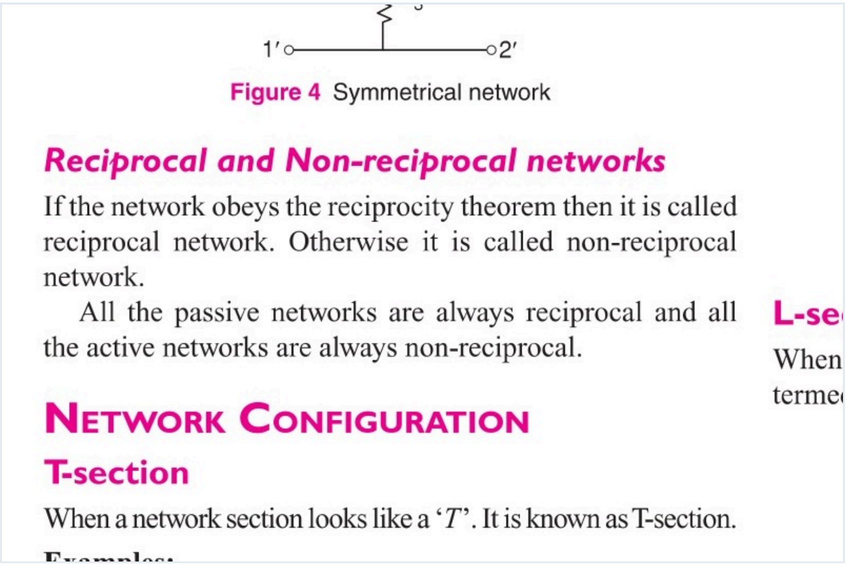


Figure 4 Symmetrical network

Reciprocal and Non-reciprocal networks

If the network obeys the reciprocity theorem then it is called reciprocal network. Otherwise it is called non-reciprocal network.

All the passive networks are always reciprocal and all the active networks are always non-reciprocal.

NETWORK CONFIGURATION

T-section

When a network section looks like a '*T*'. It is known as T-section.

L-se

When

terme

A. 12 ohm

B. 3 ohm

C. 0 ohm

D. 6 ohm

Embedded figure file: images/diagrams/electrical-flt02/q56-norton-network.jpg

Elimination Rule 9 — Resonance / X_L = X_C identity

Evidence: ESTABLISHED — 47 independent (44 bank + 3 VALID PYQ) · **Streams:** Electrical

What this rule means

At series resonance XL = XC so net reactive part is zero and Z = R (minimum). Parallel resonance is the dual (impedance maximum).

When to use

Series/parallel RLC, ω0, or impedance at resonance.

Application steps

1. Confirm series vs parallel from the figure/stem.
2. At series resonance strike any option that still carries j(XL-XC) ≠ 0.
3. Use omega0 = 1/√(LC) only when that is the ask.

```
Memory cue: Series resonance: XL = XC → Z = R
```

Core elimination move

Series resonance → Z = R (min). Parallel → Z max. Eliminate options that keep j(XL−XC) nonzero at the stated resonance.

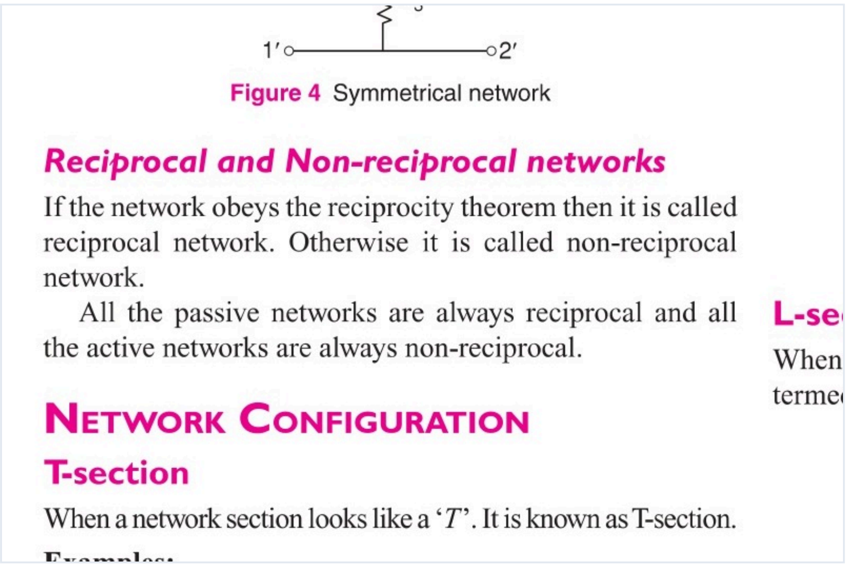
Evidence examples (sample references)

Source exam	Reference	Stem (truncated)
Electrical FLT-01	`data/electrical/ee-flt01.js#Q13`	In equal-area criterion, for a sudden load rejection on a machine, accelerating area is bounded by:
Electrical FLT-01	`data/electrical/ee-flt01.js#Q19`	A separately excited DC motor has Va=220 V, Ra=1 Ω, Ia=20 A, φ constant. If Va is halved and torque kept sa...
Electrical FLT-01	`data/electrical/ee-flt01.js#Q21`	Alternator: Vt=1 pu, Ia=1 pu, Xs=1.0 pu, Ra≈0, pf=0.8 lag. |Ef| ≈:
Electrical FLT-02	`data/electrical/ee-flt02.js#Q18`	From the alternator phasor diagram (Ia leading Vt), the machine is operating as:
Electrical ST-EC	`data/electrical/st/ee-st-ec-01.js#Q11`	Source-free series RLC underdamped ωd=:
Electrical ST-EM	`data/electrical/st/ee-st-em-01.js#Q3`	Sync machine: overexcited on infinite bus behaves as:

Figure evidence (actual diagrams — not path titles)

Electrical FLT-01 · `data/electrical/ee-flt01.js#Q32` · Key D

The two-port network shown has series arms Z1 and Z2 with a single shunt Z3 to the common return. It is best classified as:



A. π-section (two shunts, one series)

B. Lattice section with crossed arms

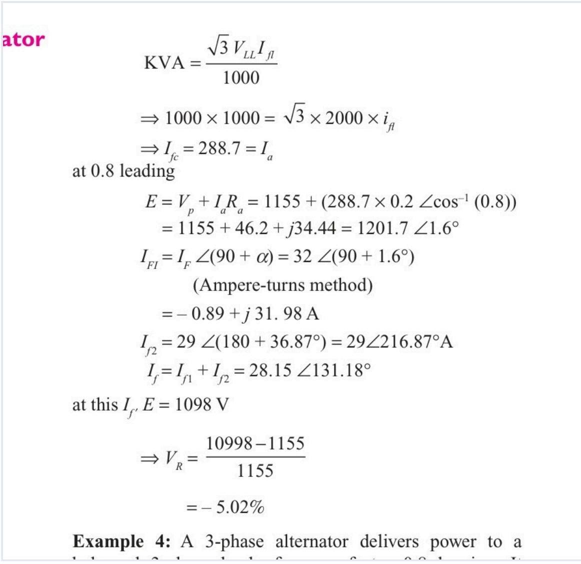
C. Symmetrical T only if Z1 = Z2; otherwise unsymmetrical T

D. Only a series RLC tank

Embedded figure file: images/diagrams/electrical-flt01/q32-t-section.jpg

Electrical FLT-02 · `data/electrical/ee-flt02.js#Q18` · Key C

From the alternator phasor diagram (Ia leading Vt), the machine is operating as:


ator

$$\text{KVA} = \frac{\sqrt{3}\,V_{LL} I_{fl}}{1000}$$

$$\Rightarrow 1000 \times 1000 = \sqrt{3} \times 2000 \times i_{fl}$$

$$\Rightarrow I_{fc} = 288.7 = I_a$$

at 0.8 leading

$$E = V_p + I_a R_a = 1155 + (288.7 \times 0.2\ \angle\cos^{-1}(0.8))$$

$$= 1155 + 46.2 + j34.44 = 1201.7\ \angle 1.6°$$

$$I_{FI} = I_F\ \angle(90 + \alpha) = 32\ \angle(90 + 1.6°)$$

(Ampere-turns method)

$$= -0.89 + j\,31.\,98\ \text{A}$$

$$I_{f2} = 29\ \angle(180 + 36.87°) = 29\angle 216.87°\text{A}$$

$$I_f = I_{f1} + I_{f2} = 28.15\ \angle 131.18°$$

at this I_f, $E = 1098$ V

$$\Rightarrow V_R = \frac{10998 - 1155}{1155}$$

$$= -5.02\%$$

Example 4: A 3-phase alternator delivers power to a

A. Short-circuit test

B. Zero power exchange

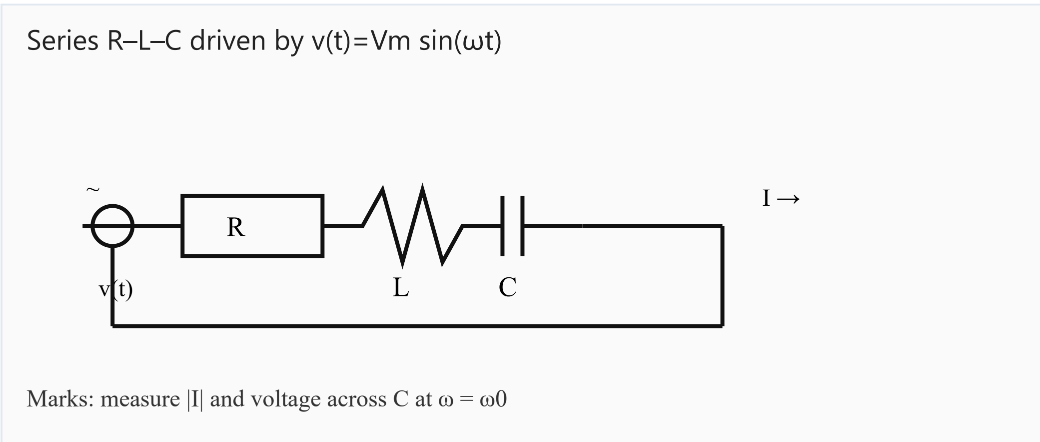
C. Leading power-factor generator

D. Lagging motor only

Embedded figure file: images/diagrams/electrical-flt02/q18-sync-phasor.jpg

Electrical FLT-02 · `data/electrical/ee-flt02.js#Q30` · Key C

Series R=10 ohm, L=20 mH, C=50 uF at resonance (rad/s), circuit impedance magnitude is:



A. 20 ohm

B. 0 ohm

C. 10 ohm

D. 50 ohm

Embedded figure file: images/diagrams/electrical-flt02/q30-series-rlc.svg

Electrical ST-EC · `data/electrical/st/ee-st-ec-01.js#Q14` · Key A

A series rlc circuit is adjusted to resonance. Which statement is correct?

Series R–L–C driven by v(t)=Vm sin(ωt)

Marks: measure |I| and voltage across C at ω = ω0

A. Its impedance is minimum and power factor is unity.

B. Its impedance is maximum and current is zero.

C. Inductive reactance exceeds capacitive reactance.

D. The supply frequency becomes zero.

Embedded figure file: images/diagrams/electrical-st-ec/q07-series-rlc.svg

Elimination Rule 10 — Sign / sense / tension–compression flip

Evidence: ESTABLISHED — 270 independent (259 bank + 11 VALID PYQ) · **Streams:** Civil, Electrical

What this rule means

Many options are numerically close but reverse tension/compression or reaction direction. Equilibrium sense is part of the answer.

When to use

Reactions, BMD jump, member force T/C, or direction in options.

Application steps

1. Fix one free-body sign convention from the figure.
2. Eliminate options with correct magnitude but wrong sense (T/C or up/down).

```
Memory cue: Sum Fx = 0, Sum Fy = 0, Sum M = 0 (sense matters)
```

Core elimination move

Fix one free-body sign convention from the figure. Eliminate options whose magnitude is right but sense (↑/↓, T/C) contradicts equilibrium.

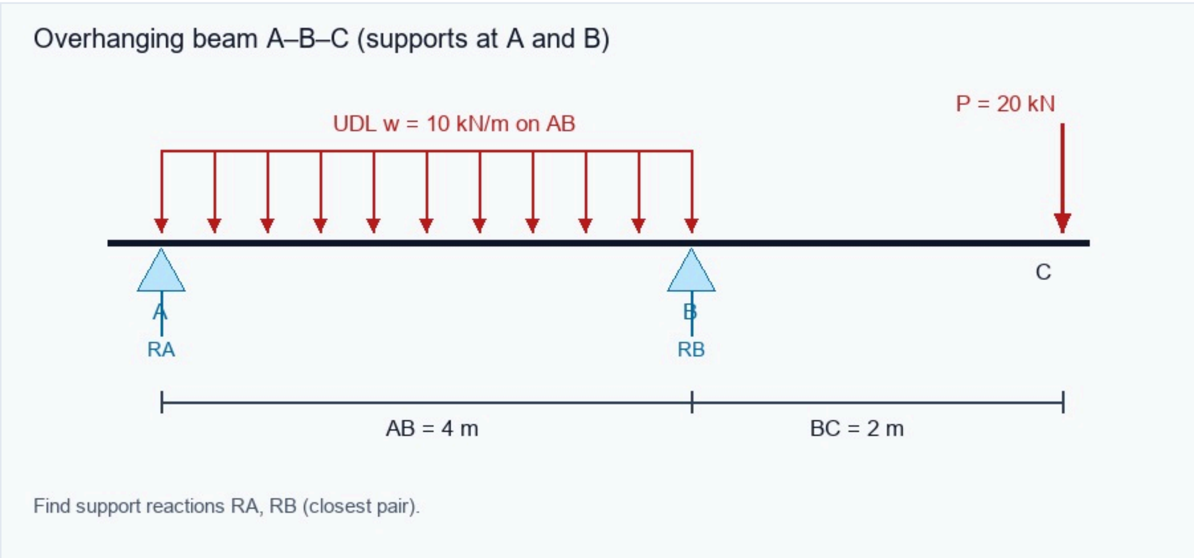
Evidence examples (sample references)

Source exam	Reference	Stem (truncated)
Civil FLT-01	`data/civil/ce-flt01.js#Q1`	Following a gantry-tie inspection at a substation, the engineer records a 20 mm diameter member under 55 kN...
Civil FLT-01	`data/civil/ce-flt01.js#Q3`	Proof-load planning uses an overhanging beam A–B–C with supports at A and B: AB = 4 m, overhang BC = 2 m, U...
Civil FLT-01	`data/civil/ce-flt01.js#Q7`	During review of a stepped shaft ABC, an end torque of 1000 N·m is applied at C while B is an intermediate ...
Civil FLT-02	`data/civil/ce-flt02.js#Q1`	A 2.0 m long steel tie of uniform 500 mm^2 area carries a gradually applied 50 kN service load. Neglecting s...
Civil ST-FE	`data/civil/st/ce-st-fe-01.js#Q4`	A pile group contains 4 piles. Each pile has ultimate base resistance 600 kPa over toe area 0.2 m^2; shaft r...
Civil ST-FM	`data/civil/st/ce-st-fm-01.js#Q2`	Water flows in a 0.2 m diameter pipe at mean velocity 4 m/s. Discharge is closest to:

Figure evidence (actual diagrams — not path titles)

Civil FLT-01 · `data/civil/ce-flt01.js#Q3` · Key A

Proof-load planning uses an overhanging beam A–B–C with supports at A and B: AB = 4 m, overhang BC = 2 m, UDL w = 10 kN/m over AB, and a tip load P = 20 kN at C (also shown on the figure). Which support-reaction pair (RA



A. 10 kN, 50 kN

B. 15 kN, 45 kN

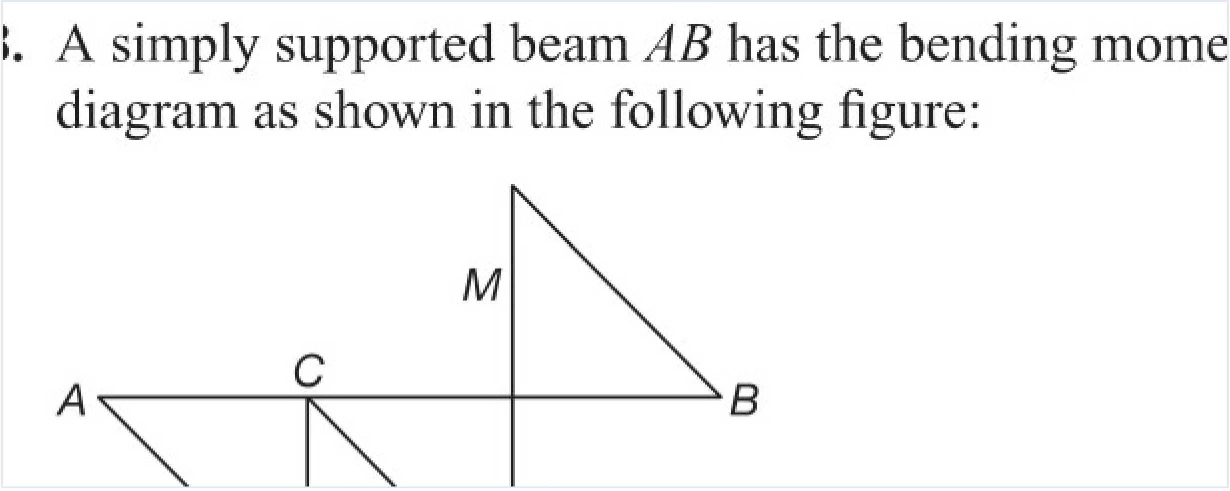
C. 20 kN, 40 kN

D. 25 kN, 35 kN

Embedded figure file: images/diagrams/civil-flt01/q03-overhang-reactions.jpg

Civil FLT-01 · `data/civil/ce-flt01.js#Q9` · Key A

On the plotted bending-moment diagram, a marked vertical jump occurs at section C while shear remains finite on both sides of C and no local section change is recorded (jump labelled at C on the figure). Which loading in



A. A concentrated couple acts at C

B. A point load acts at C

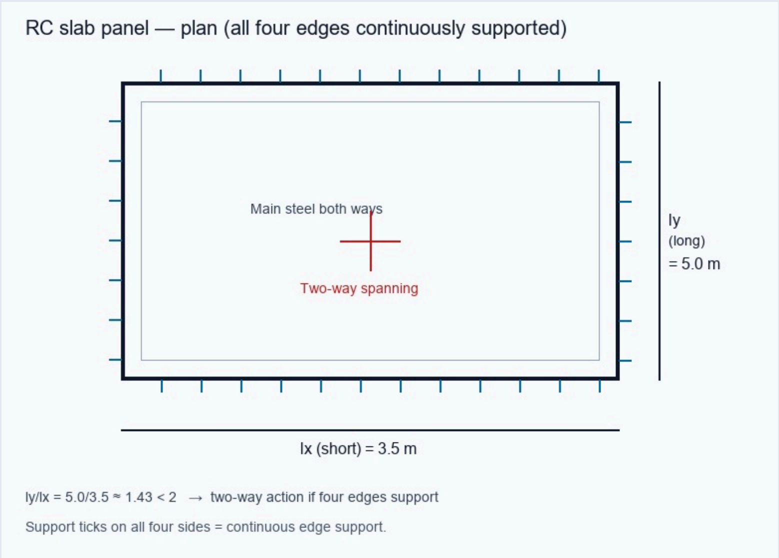
C. A UDL starts at C

D. The flexural rigidity changes at C

Embedded figure file: images/diagrams/civil-flt01/q09-bmd-jump.jpg

Civil FLT-01 · `data/civil/ce-flt01.js#Q18` · Key C

An RC slab panel has clear spans lx = 3.5 m (short) and ly = 5.0 m (long), so ly/lx ≈ 1.43, and all four edges are continuously supported (dimensions and edge-support ticks also shown on the figure). Assertion: the panel



A. Both true; R does not explain A

B. A true, R false

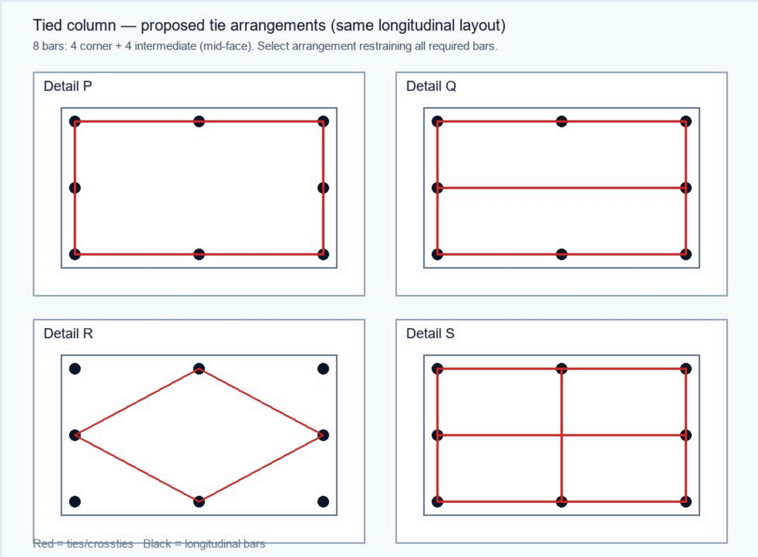
C. Both true; R explains A

D. A false, R true

Embedded figure file: images/diagrams/civil-flt01/q18-slab-supports.jpg

Civil FLT-01 · `data/civil/ce-flt01.js#Q21` · Key D

At reinforcement inspection, a tied column shows the longitudinal-bar layout with intermediate bars on each face, and four proposed tie arrangements labelled Detail P, Q, R and S on the figure. Which marked arrangement b



A. Detail P

B. Detail Q

C. Detail R

D. Detail S

Embedded figure file: images/diagrams/civil-flt01/q21-column-tie-detail.jpg

Elimination Rule 11 — Figure-dependency: cover the figure test

Evidence: ESTABLISHED — 304 independent (109 bank + 195 VALID PYQ) · **Streams:** Civil, Electrical

What this rule means

If a figure is attached, the keyed answer usually depends on a labelled length, support, or curve. Solving from the stem alone is a dependency failure.

When to use

Stem says ‘shown’, ‘figure’, or options name Detail/Curve labels.

Application steps

1. Read every dimension and support on the crop before picking.
2. Strike options that ignore a labelled overhang, load, or curve name.
3. If the stem names Detail/Curve labels, those labels must appear on the figure.

`Memory cue: Cover-the-figure test: cannot answer correctly with figure hidden`

Core elimination move

If you can pick the answer without the figure, stop — stem/figure mismatch. Otherwise eliminate options that ignore a labelled dimension, support, or curve on the crop.

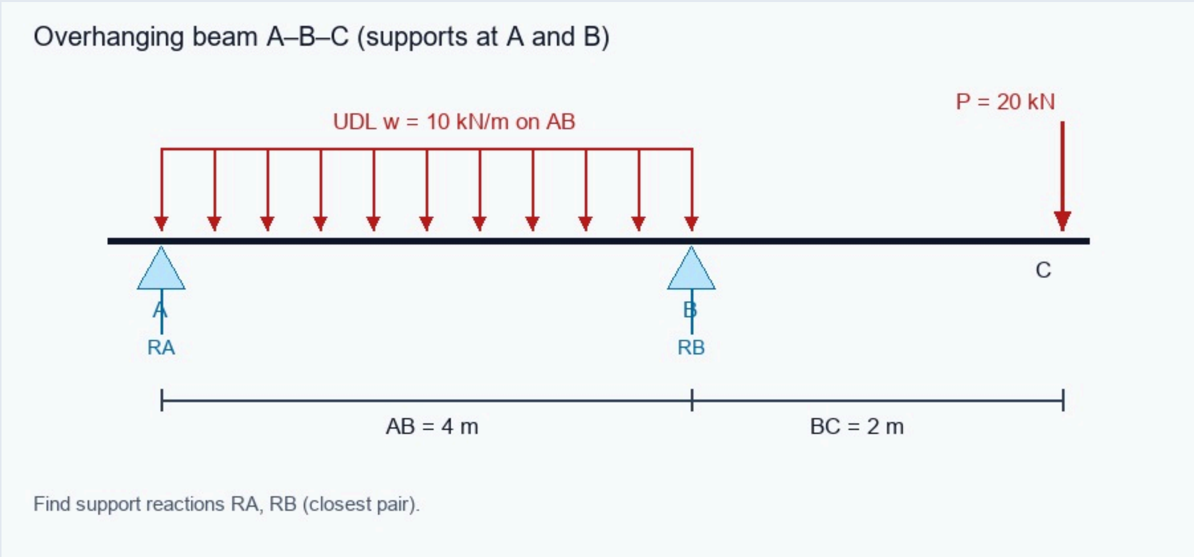
Evidence examples (sample references)

Source exam	Reference	Stem (truncated)
Civil FLT-01	`data/civil/ce-flt01.js#Q3`	Proof-load planning uses an overhanging beam A–B–C with supports at A and B: AB = 4 m, overhang BC = 2 m, U…
Civil FLT-01	`data/civil/ce-flt01.js#Q9`	On the plotted bending-moment diagram, a marked vertical jump occurs at section C while shear remains finit…
Civil FLT-01	`data/civil/ce-flt01.js#Q18`	An RC slab panel has clear spans lx = 3.5 m (short) and ly = 5.0 m (long), so ly/lx ≈ 1.43, and all four ed…
Civil FLT-02	`data/civil/ce-flt02.js#Q3`	The overhanging beam in the crop carries the shown point load and UDL; span lengths and load locations appe…
Civil ST-FE	`data/civil/st/ce-st-fe-01.js#Q3`	The shown square footing distributes a concentric service load of 260 kN over a 2.3 m × 2.3 m base. The ave…
Civil ST-FM	`data/civil/st/ce-st-fm-01.js#Q3`	The shown venturimeter is installed in a 0.2 m diameter inlet pipe. If the inlet velocity is 3 m/s, the dis…

Figure evidence (actual diagrams — not path titles)

Civil FLT-01 · `data/civil/ce-flt01.js#Q3` · Key A

Proof-load planning uses an overhanging beam A–B–C with supports at A and B: AB = 4 m, overhang BC = 2 m, UDL w = 10 kN/m over AB, and a tip load P = 20 kN at C (also shown on the figure). Which support-reaction pair (RA



A. 10 kN, 50 kN

B. 15 kN, 45 kN

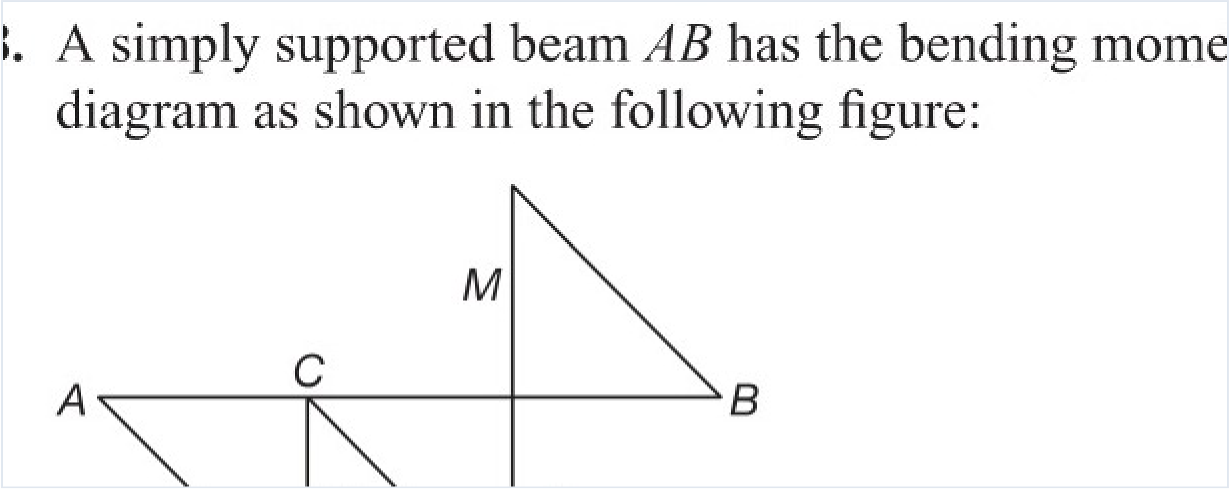
C. 20 kN, 40 kN

D. 25 kN, 35 kN

Embedded figure file: images/diagrams/civil-flt01/q03-overhang-reactions.jpg

Civil FLT-01 · `data/civil/ce-flt01.js#Q9` · Key A

On the plotted bending-moment diagram, a marked vertical jump occurs at section C while shear remains finite on both sides of C and no local section change is recorded (jump labelled at C on the figure). Which loading in


A simply supported beam AB has the bending mome diagram as shown in the following figure:

A. A concentrated couple acts at C

B. A point load acts at C

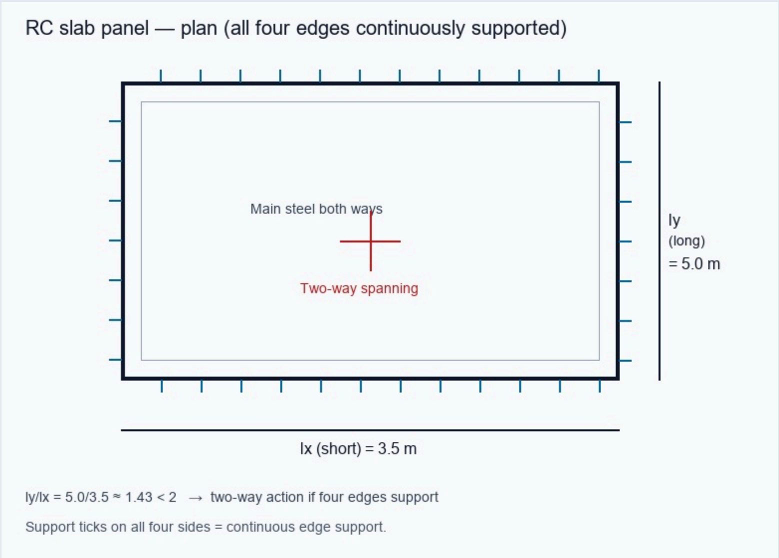
C. A UDL starts at C

D. The flexural rigidity changes at C

Embedded figure file: images/diagrams/civil-flt01/q09-bmd-jump.jpg

Civil FLT-01 · `data/civil/ce-flt01.js#Q18` · Key C

An RC slab panel has clear spans lx = 3.5 m (short) and ly = 5.0 m (long), so ly/lx ≈ 1.43, and all four edges are continuously supported (dimensions and edge-support ticks also shown on the figure). Assertion: the panel



A. Both true; R does not explain A

B. A true, R false

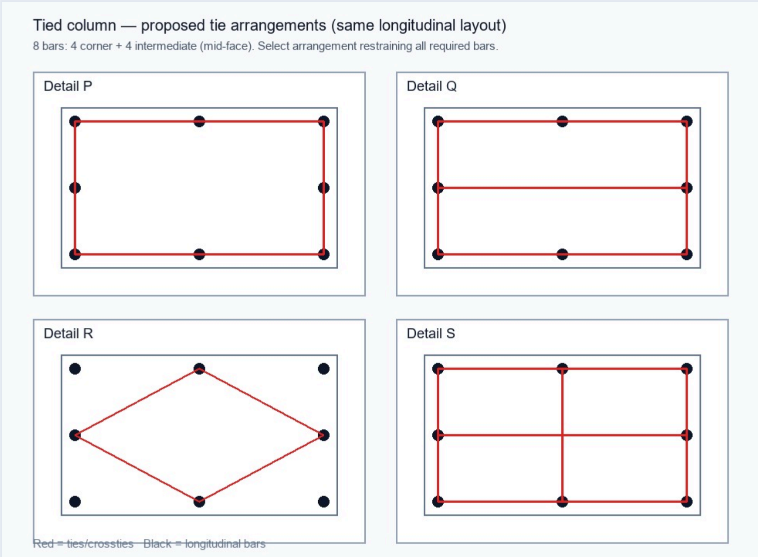
C. Both true; R explains A

D. A false, R true

Embedded figure file: images/diagrams/civil-flt01/q18-slab-supports.jpg

Civil FLT-01 · `data/civil/ce-flt01.js#Q21` · Key D

At reinforcement inspection, a tied column shows the longitudinal-bar layout with intermediate bars on each face, and four proposed tie arrangements labelled Detail P, Q, R and S on the figure. Which marked arrangement b



A. Detail P

B. Detail Q

C. Detail R

D. Detail S

Embedded figure file: images/diagrams/civil-flt01/q21-column-tie-detail.jpg

Elimination Rule 12 — Rankine / Terzaghi / earth-pressure term drop

Evidence: ESTABLISHED — 86 independent (85 bank + 1 VALID PYQ) · **Streams:** Civil

What this rule means

Bearing capacity and earth-pressure formulae have multiple additive terms. Dropping Nγ, swapping Ka/Kp, or using dry unit weight under water are classic traps.

When to use

Bearing capacity or lateral earth pressure with Nc, Nq, Nγ or Ka/Kp.

Application steps

1. Write the full Terzaghi or Rankine expression used in the stem.
2. Strike options that omit a term or use Ka where Kp is required (or vice versa).
3. If submerged, use buoyant unit weight as required.

```
Memory cue: qu = c Nc + q Nq + 0.5 γ B Nγ (Terzaghi form)
```

Core elimination move

Write the full Terzaghi/Rankine expression. Eliminate options that drop the $\gamma B N$ term, swap $K_a \leftrightarrow K_p$, or use dry γ when stem says submerged.

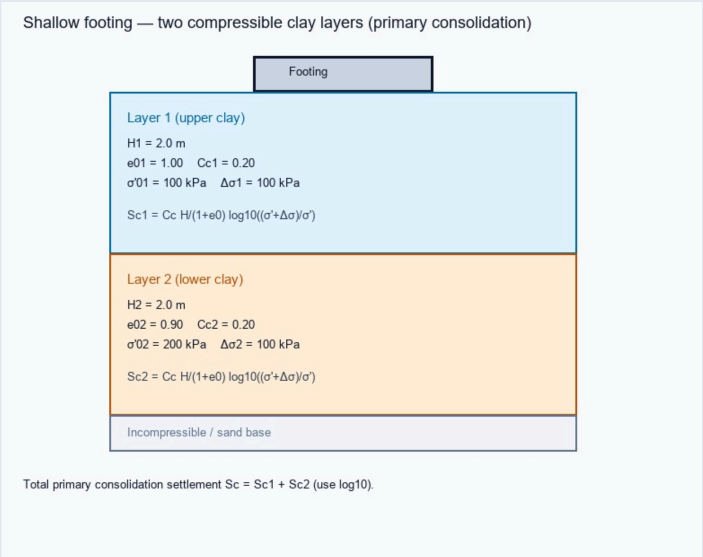
Evidence examples (sample references)

Source exam	Reference	Stem (truncated)
Civil FLT-01	`data/civil/ce-flt01.js#Q11`	During buckling checks for a slender pin-ended strut, a junior engineer proposes to use the classical Euler…
Civil FLT-01	`data/civil/ce-flt01.js#Q38`	At pump inspection during a commissioning hold, crackling noise, fluctuating delivery head and pitting mark…
Civil FLT-01	`data/civil/ce-flt01.js#Q53`	Primary consolidation under a shallow footing uses two clay layers (also tabulated on the figure): Layer 1 …
Civil FLT-02	`data/civil/ce-flt02.js#Q9`	A simply supported deep timber beam develops horizontal cracks near the neutral axis although bending stres…
Civil ST-FE	`data/civil/st/ce-st-fe-01.js#Q1`	For a strip footing, take c = 20 kPa, Nc = 25, q = 30 kPa, Nq = 12, γ = 18 kN/m³, B = 2 m and Nγ = 8. Terza…
Civil ST-RCC	`data/civil/st/ce-st-rcc-01.js#Q5`	A square footing 2.1 m by 2.1 m carries a service column load of 190 kN (self-weight ignored). Average soil…

Figure evidence (actual diagrams — not path titles)

Civil FLT-01 · `data/civil/ce-flt01.js#Q53` · Key D

Primary consolidation under a shallow footing uses two clay layers (also tabulated on the figure): Layer 1 — H1 = 2.0 m, e01 = 1.00, Cc1 = 0.20, σ′01 = 100 kPa, Δσ1 = 100 kPa; Layer 2 — H2 = 2.0 m, e02 = 0.90, Cc2 = 0.20



A. 42 mm

B. 68 mm

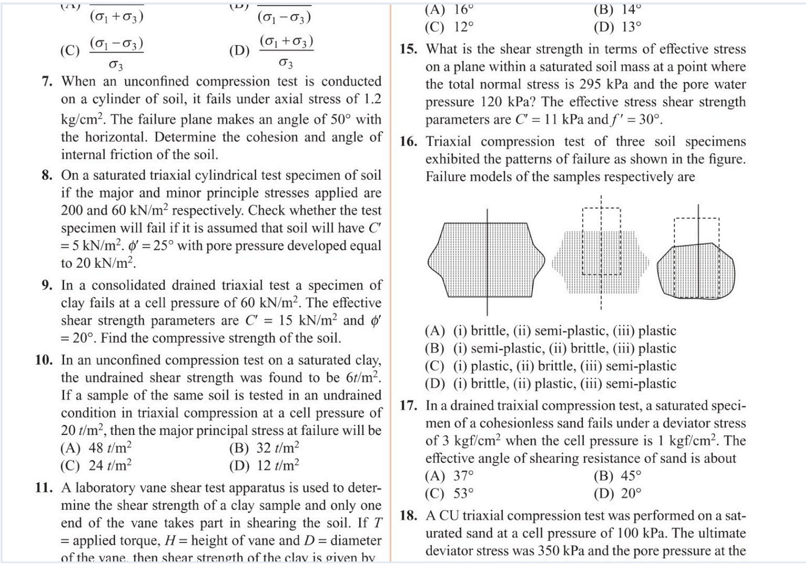
C. 126 mm

D. 95 mm

Embedded figure file: images/diagrams/civil-flt01/q53-layered-footing.jpg

Civil FLT-02 · `data/civil/ce-flt02.js#Q52` · Key B

A footing is called “compensated” when excavation removes soil weight that offsets part of the structural pressure. What is the principal benefit?



(A) $\frac{\quad}{(\sigma_1+\sigma_3)}$ (B) $\frac{\quad}{(\sigma_1-\sigma_3)}$

(C) $\frac{(\sigma_1-\sigma_3)}{\sigma_3}$ (D) $\frac{(\sigma_1+\sigma_3)}{\sigma_3}$

7. When an unconfined compression test is conducted on a cylinder of soil, it fails under axial stress of 1.2 kg/cm^2. The failure plane makes an angle of 50° with the horizontal. Determine the cohesion and angle of internal friction of the soil.

8. On a saturated triaxial cylindrical test specimen of soil if the major and minor principle stresses applied are 200 and 60 kN/m^2 respectively. Check whether the test specimen will fail if it is assumed that soil will have $C' = 5$ kN/m^2. $\phi' = 25°$ with pore pressure developed equal to 20 kN/m^2.

9. In a consolidated drained triaxial test a specimen of clay fails at a cell pressure of 60 kN/m^2. The effective shear strength parameters are $C' = 15$ kN/m^2 and $\phi' = 20°$. Find the compressive strength of the soil.

10. In an unconfined compression test on a saturated clay, the undrained shear strength was found to be 6t/m^2. If a sample of the same soil is tested in an undrained condition in triaxial compression at a cell pressure of 20 t/m^2, then the major principal stress at failure will be

(A) 48 t/m^2 (B) 32 t/m^2
(C) 24 t/m^2 (D) 12 t/m^2

11. A laboratory vane shear test apparatus is used to determine the shear strength of a clay sample and only one end of the vane takes part in shearing the soil. If T = applied torque, H = height of vane and D = diameter of the vane, then shear strength of the clay is given by

(A) 16° (B) 14°
(C) 12° (D) 13°

15. What is the shear strength in terms of effective stress on a plane within a saturated soil mass at a point where the total normal stress is 295 kPa and the pore water pressure 120 kPa? The effective stress shear strength parameters are $C' = 11$ kPa and $f' = 30°$.

16. Triaxial compression test of three soil specimens exhibited the patterns of failure as shown in the figure. Failure models of the samples respectively are

(A) (i) brittle, (ii) semi-plastic, (iii) plastic
(B) (i) semi-plastic, (ii) brittle, (iii) plastic
(C) (i) plastic, (ii) brittle, (iii) semi-plastic
(D) (i) brittle, (ii) plastic, (iii) semi-plastic

17. In a drained traixial compression test, a saturated specimen of a cohesionless sand fails under a deviator stress of 3 kgf/cm^2 when the cell pressure is 1 kgf/cm^2. The effective angle of shearing resistance of sand is about

(A) 37° (B) 45°
(C) 53° (D) 20°

18. A CU triaxial compression test was performed on a saturated sand at a cell pressure of 100 kPa. The ultimate deviator stress was 350 kPa and the pore pressure at the

A. zero gross pressure

B. reduced net foundation pressure and settlement demand

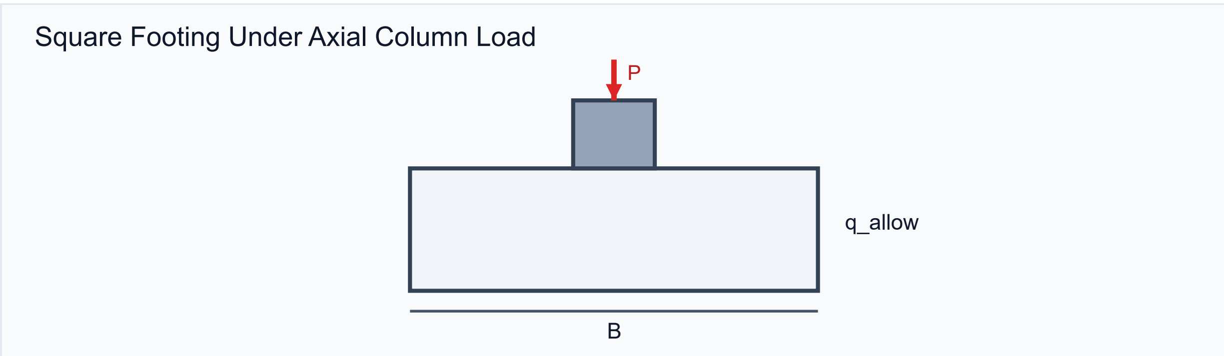
C. no need for bearing check

D. guaranteed uplift resistance

Embedded figure file: images/diagrams/civil-flt02/q52-portal-frame.jpg

Civil ST-FE · `data/civil/st/ce-st-fe-01.js#Q3` · Key D

The shown square footing distributes a concentric service load of 260 kN over a 2.3 m × 2.3 m base. The average contact pressure is closest to:



A. 39.32 kPa

B. 58.98 kPa

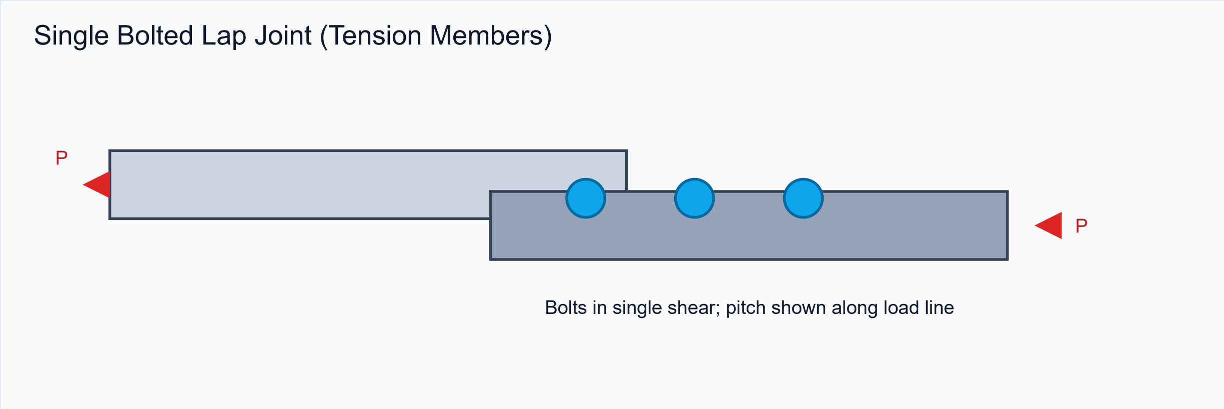
C. 73.72 kPa

D. 49.15 kPa

Embedded figure file: images/diagrams/civil-st-fe/foundation-square-footing.svg

Civil ST-STEEL · `data/civil/st/ce-st-steel-01.js#Q3` · Key D

The shown bolted lap joint transfers the load through 4 identical bolts. If each bolt has governing design capacity 45 kN, the concentric joint capacity is closest to:



A. 144 kN

B. 216 kN

C. 270 kN

D. 180 kN

Embedded figure file: images/diagrams/civil-st-steel/steel-bolted-lap.svg

Elimination Rule 13 — Mohr / principal-stress pair consistency

Evidence: ESTABLISHED — 26 independent (23 bank + 3 VALID PYQ) · **Streams:** Civil

What this rule means

Principal stresses from Mohr's circle must satisfy $\sigma 1 + \sigma 2 = \sigma x + \sigma y$. Treating shear as a normal stress breaks that invariant.

When to use

σx, σy, τxy given; options are $\sigma 1$, $\sigma 2$, or θp.

Application steps

1. Compute σ_avg and R; then σ1,2 = σ_avg ± R.
2. Strike pairs that violate σ1+σ2 = σx+σy.

```
Memory cue: σ1,2 = (σx+σy)/2 ± √[((σx-σy)/2)^2 + τ^2]
```

Core elimination move

$\sigma avg = (\sigma x+\sigma y)/2$; $R = \sqrt{[((\sigma x-\sigma y)/2)^2+\tau^2]}$. Eliminate pairs that violate $\sigma 1+\sigma 2 = \sigma x+\sigma y$ or that treat τ as a normal stress.

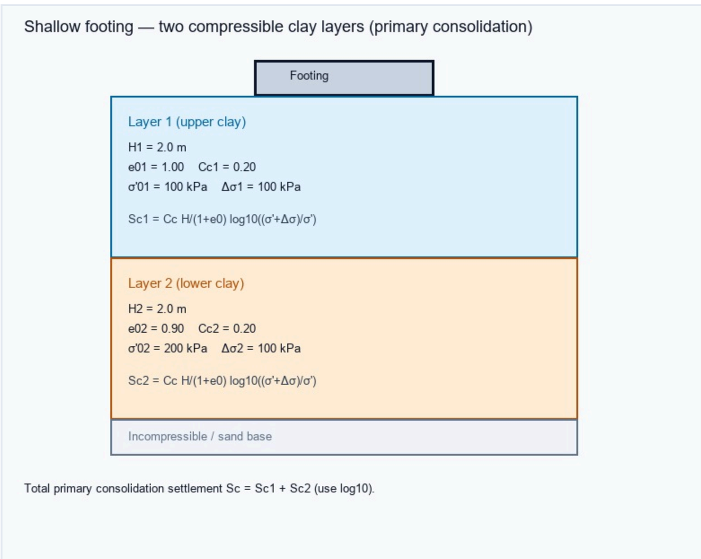
Evidence examples (sample references)

Source exam	Reference	Stem (truncated)
Civil FLT-01	`data/civil/ce-flt01.js#Q5`	While interpreting a strain-rosette investigation on a critical plane-stress element, the engineer reviews ...
Civil FLT-01	`data/civil/ce-flt01.js#Q43`	A built-up column reviewed in the design office has different effective lengths and different radii of gyra...
Civil FLT-01	`data/civil/ce-flt01.js#Q51`	While checking an inclined-roof purlin under combined gravity and wind-reversal actions, the resultant load...
Civil FLT-02	`data/civil/ce-flt02.js#Q12`	At a point, \(\sigma_x=80\) MPa tension, \(\sigma_y=20\) MPa compression and \(\tau_{xy}=30\) MPa. Determin...
Civil ST-RCC	`data/civil/st/ce-st-rcc-01.js#Q4`	A 300 mm × 300 mm short tied column carries factored axial load 1050 kN. The average factored stress Pu/Ag ...
Civil ST-SOM	`data/civil/st/ce-st-som-01.js#Q13`	At a point in a plate, σx = 92 MPa, σy = 22 MPa and τxy = 30 MPa. The major principal stress is closest to:

Figure evidence (actual diagrams — not path titles)

Civil FLT-01 · `data/civil/ce-flt01.js#Q53` · Key D

Primary consolidation under a shallow footing uses two clay layers (also tabulated on the figure): Layer 1 — H1 = 2.0 m, e01 = 1.00, Cc1 = 0.20, σ′01 = 100 kPa, Δσ1 = 100 kPa; Layer 2 — H2 = 2.0 m, e02 = 0.90, Cc2 = 0.20



A. 42 mm

B. 68 mm

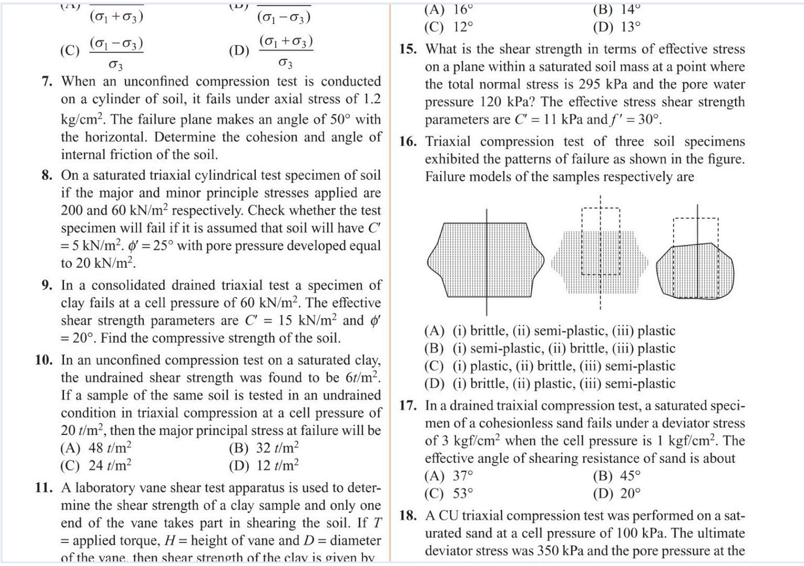
C. 126 mm

D. 95 mm

Embedded figure file: images/diagrams/civil-flt01/q53-layered-footing.jpg

Civil FLT-02 · `data/civil/ce-flt02.js#Q52` · Key B

A footing is called “compensated” when excavation removes soil weight that offsets part of the structural pressure. What is the principal benefit?



(A) $\overline{(\sigma_1+\sigma_3)}$ (B) $\overline{(\sigma_1-\sigma_3)}$

(C) $\dfrac{(\sigma_1-\sigma_3)}{\sigma_3}$ (D) $\dfrac{(\sigma_1+\sigma_3)}{\sigma_3}$

7. When an unconfined compression test is conducted on a cylinder of soil, it fails under axial stress of 1.2 kg/cm^2. The failure plane makes an angle of 50° with the horizontal. Determine the cohesion and angle of internal friction of the soil.

8. On a saturated triaxial cylindrical test specimen of soil if the major and minor principle stresses applied are 200 and 60 kN/m^2 respectively. Check whether the test specimen will fail if it is assumed that soil will have C' = 5 kN/m^2. ϕ' = 25° with pore pressure developed equal to 20 kN/m^2.

9. In a consolidated drained triaxial test a specimen of clay fails at a cell pressure of 60 kN/m^2. The effective shear strength parameters are C' = 15 kN/m^2 and ϕ' = 20°. Find the compressive strength of the soil.

10. In an unconfined compression test on a saturated clay, the undrained shear strength was found to be 6t/m^2. If a sample of the same soil is tested in an undrained condition in triaxial compression at a cell pressure of 20 t/m^2, then the major principal stress at failure will be
(A) 48 t/m^2 (B) 32 t/m^2
(C) 24 t/m^2 (D) 12 t/m^2

11. A laboratory vane shear test apparatus is used to determine the shear strength of a clay sample and only one end of the vane takes part in shearing the soil. If T = applied torque, H = height of vane and D = diameter of the vane, then shear strength of the clay is given by

(A) 16° (B) 14°
(C) 12° (D) 13°

15. What is the shear strength in terms of effective stress on a plane within a saturated soil mass at a point where the total normal stress is 295 kPa and the pore water pressure 120 kPa? The effective stress shear strength parameters are C' = 11 kPa and f' = 30°.

16. Triaxial compression test of three soil specimens exhibited the patterns of failure as shown in the figure. Failure models of the samples respectively are

(A) (i) brittle, (ii) semi-plastic, (iii) plastic
(B) (i) semi-plastic, (ii) brittle, (iii) plastic
(C) (i) plastic, (ii) brittle, (iii) semi-plastic
(D) (i) brittle, (ii) plastic, (iii) semi-plastic

17. In a drained traixial compression test, a saturated specimen of a cohesionless sand fails under a deviator stress of 3 kgf/cm^2 when the cell pressure is 1 kgf/cm^2. The effective angle of shearing resistance of sand is about
(A) 37° (B) 45°
(C) 53° (D) 20°

18. A CU triaxial compression test was performed on a saturated sand at a cell pressure of 100 kPa. The ultimate deviator stress was 350 kPa and the pore pressure at the

A. zero gross pressure

B. reduced net foundation pressure and settlement demand

C. no need for bearing check

D. guaranteed uplift resistance

Embedded figure file: images/diagrams/civil-flt02/q52-portal-frame.jpg

Elimination Rule 14 — SFD↔BMD jump / couple vs point-load confusion

Evidence: ESTABLISHED — 39 independent (34 bank + 5 VALID PYQ) · **Streams:** Civil

What this rule means

On SFDs/BMDs: a point load causes a shear jump; a couple causes a moment jump (no shear jump).

When to use

Beam figure with point load, UDL, or couple; options describe jumps.

Application steps

1. Identify whether the singularity is a force or a couple.
2. Strike options that swap shear-jump and moment-jump rules.

```
Memory cue: Point load → ΔV ; Couple → ΔM
```

Core elimination move

Point load → shear jump; couple → moment jump (no shear jump). Eliminate options that swap those rules.

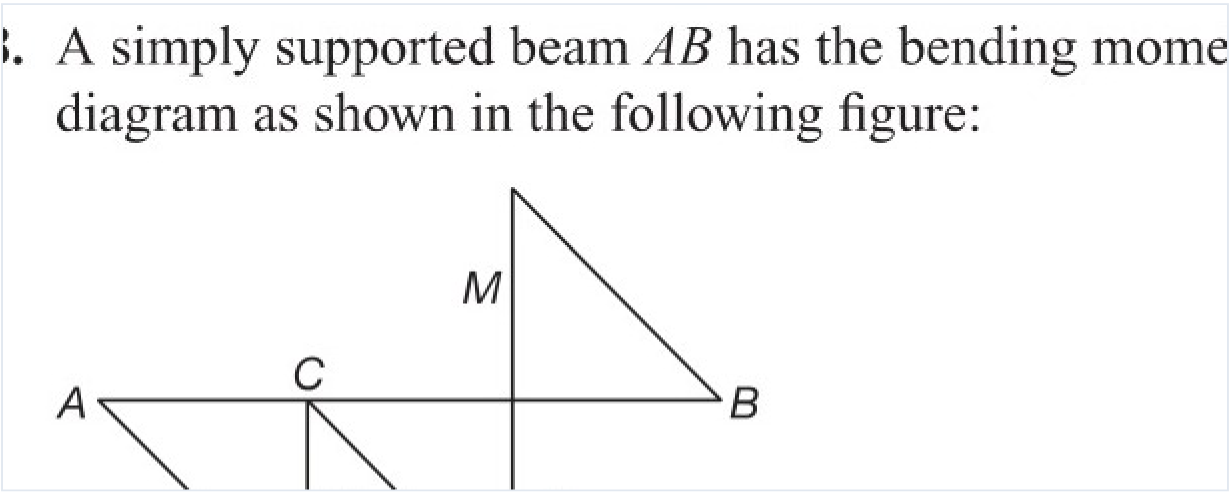
Evidence examples (sample references)

Source exam	Reference	Stem (truncated)
Civil FLT-01	`data/civil/ce-flt01.js#Q9`	On the plotted bending-moment diagram, a marked vertical jump occurs at section C while shear remains finit...
Civil FLT-01	`data/civil/ce-flt01.js#Q24`	In design review, factored moment is 160 kN·m while the limiting singly reinforced capacity of the fixed be...
Civil FLT-01	`data/civil/ce-flt01.js#Q25`	A design review finds that the factored moment exceeds the limiting capacity of the same section as a singl...
Civil FLT-02	`data/civil/ce-flt02.js#Q14`	Match loading change with the qualitative SFD/BMD consequence: (1) point load, (2) applied couple, (3) UDL,...
Civil ST-RCC	`data/civil/st/ce-st-rcc-01.js#Q1`	An RCC beam of M25 and Fe500 has factored Mu = 180 kN m, b = 230 mm and d = 450 mm. Using z = 0.9d, the req...
Civil ST-SOM	`data/civil/st/ce-st-som-01.js#Q2`	A simply supported beam of span 5 m carries a uniform load of 13 kN/m over the full span. The maximum bendi...

Figure evidence (actual diagrams — not path titles)

Civil FLT-01 · `data/civil/ce-flt01.js#Q9` · Key A

On the plotted bending-moment diagram, a marked vertical jump occurs at section C while shear remains finite on both sides of C and no local section change is recorded (jump labelled at C on the figure). Which loading in



A. A concentrated couple acts at C

B. A point load acts at C

C. A UDL starts at C

D. The flexural rigidity changes at C

Embedded figure file: images/diagrams/civil-flt01/q09-bmd-jump.jpg

Elimination Rule 15 — SCR / PE latching vs holding vs firing

Evidence: ESTABLISHED — 18 independent (14 bank + 4 VALID PYQ) · **Streams:** Electrical

What this rule means

For SCRs, latching current is larger than holding current. Latching is needed to establish conduction after the gate pulse; holding keeps it on.

When to use

Thyristor/SCR gate, latching, holding, or commutation options.

Application steps

1. Remember IL > IH.
2. Strike options that swap latching with holding or that size gate width from holding alone.

```
Memory cue: Latching current > holding current
```

Core elimination move

Latching > holding; latching is the current to *establish* conduction after gate pulse. Eliminate options that swap latching↔holding or that set gate width from holding current alone.

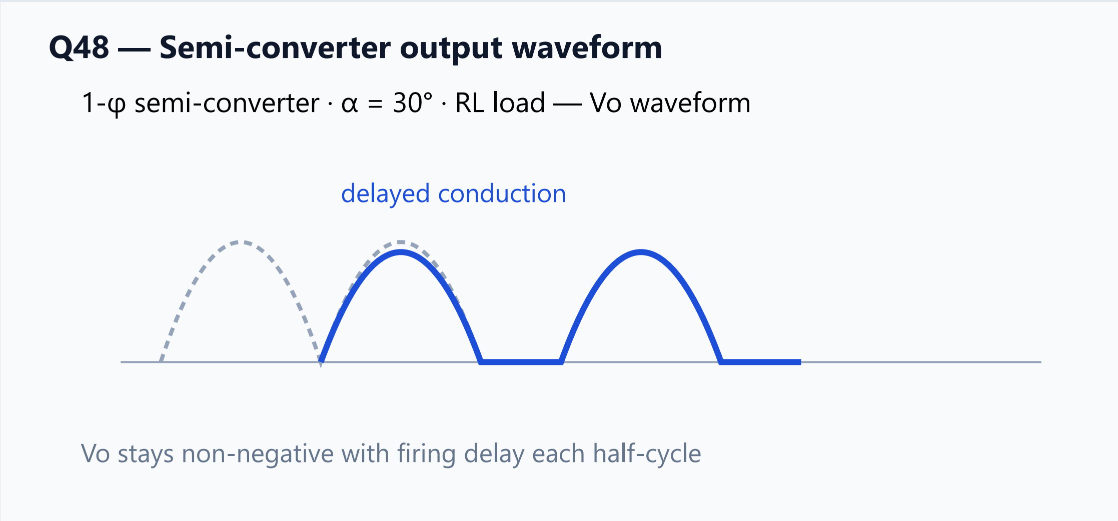
Evidence examples (sample references)

Source exam	Reference	Stem (truncated)
Electrical FLT-01	`data/electrical/ee-flt01.js#Q47`	An SCR turns off when which condition is met for sufficient time?
Electrical FLT-02	`data/electrical/ee-flt02.js#Q47`	After latch, SCR stays on until anode current falls below:
Electrical FLT-02	`data/electrical/ee-flt02.js#Q53`	In field, class-F (line) commutation of SCR succeeds when:
Electrical ST-PE	`data/electrical/st/ee-st-pe-01.js#Q1`	SCR latching current is:
VALID PYQ · 2026EE.pdf	`PYQ:2026EE.pdf#Q28`	Consider the circuit shown in Figure
VALID PYQ · APEPDCL-Assistant-Engineer-2019-Official-Paper.pdf	`PYQ:APEPDCL-Assistant-Engineer-2019-Official-Paper.pdf#Q77`	For the converter shown below, firing angle for thyristors `T1` and `T2` is `π/6`; firing angle for thyrist…

Figure evidence (actual diagrams — not path titles)

Electrical ST-PE · `data/electrical/st/ee-st-pe-01.js#Q1` · Key C

SCR latching current is:



A. Ig=0

B. gate peak

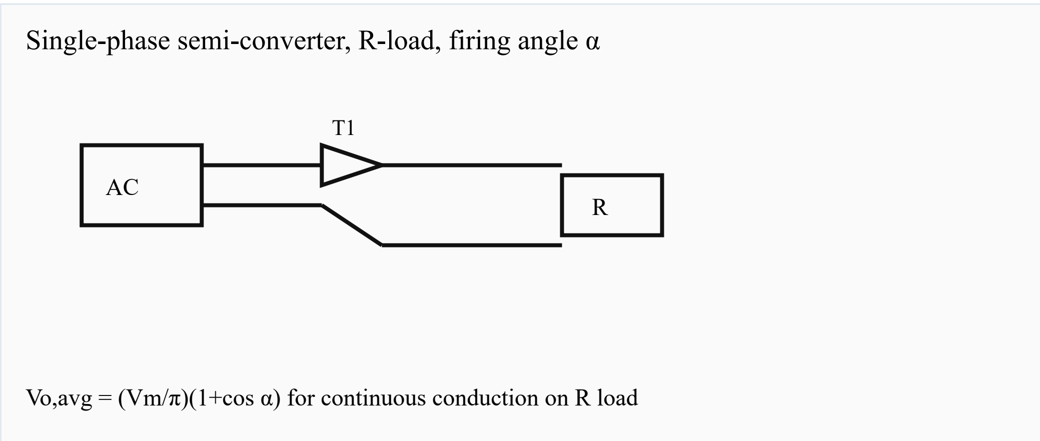
C. min on-current to keep latch after gate removed

D. holding only identical always

Embedded figure file: images/diagrams/electrical-st-pe/pe3.svg

Electrical ST-PE · `data/electrical/st/ee-st-pe-01.js#Q8` · Key A

Commutation in class B (self) SCR circuit uses:



A. LC resonating current reverse

B. only fuse

C. only relay

D. only line AC

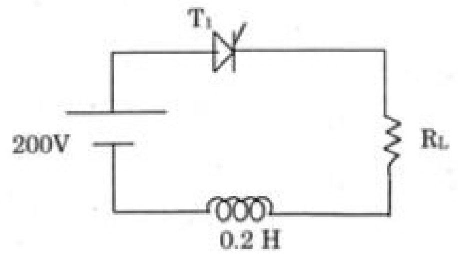
Embedded figure file: images/diagrams/electrical-st-pe/q07-single-phase-semi.svg

VALID PYQ · AP-GENCO-Tech-2012.pdf · `PYQ:AP-GENCO-Tech-2012.pdf#Q10`

The latching current of T1 is 1 mA. The minimum width of gate pulse required to turn on SCR is

Code D

10.



The latching current of T_1 is 1 mA. The minimum width of gate pulse required to turn on SCR is

(A) 2 μsec (B) 1 μsec (C) 0.5 μsec (D) 1.5 μsec

A. 2 µsec

B. 1 µsec

C. 0.5 µsec

D. 1.5 µsec

Embedded figure file: exports/PYQ_VALID_DIAGRAM_CROPS/AP-GENCO-Tech-2012/AP-GENCO-Tech-2012__p04__Q010.jpg

Elimination Rule 16 — Non-core: option that abandons the asked operation

Evidence: ESTABLISHED — 37 independent (37 bank + 0 VALID PYQ) · **Streams:** Non-core

What this rule means

Non-core items often include one option that solves a different operation (CI instead of SI, active instead of passive, wrong percentage base).

When to use

Quant / reasoning / English — one option changes the operation (%, SI, tense, code).

Application steps

1. Re-state the exact ask in one short sentence.
2. Strike any option that changes the operation.

```
Memory cue: Match the asked operation exactly (%, SI, tense, code rule)
```

Core elimination move

Re-state the exact ask (20% of 450; SI for 2 years; passive of present perfect). Eliminate options that solve a different operation (CI instead of SI, active instead of passive).

Evidence examples (sample references)

Source exam	Reference	Stem (truncated)
Civil FLT-01	`data/civil/ce-flt01.js#Q72`	A transmission-estimate package rises from ₹8.0 crore to ₹9.2 crore after a design revision. The percentage...
Civil FLT-01	`data/civil/ce-flt01.js#Q76`	A mobilisation advance of ₹5,00,000 is charged simple interest at 12% per annum for 8 months. The interest ...
Civil FLT-01	`data/civil/ce-flt01.js#Q77`	A contractor's quoted cost for a bay package is ₹12 lakh. He allows 10% discount on the quote and still mak...
Civil FLT-02	`data/civil/ce-flt02.js#Q73`	A price rises 20%, falls 10%, then attracts 5% tax. Relative to the original pre-tax price, the final amoun...
Electrical FLT-01	`data/electrical/ee-flt01.js#Q72`	A transmission-estimate package rises from ₹8.0 crore to ₹9.2 crore after a design revision. The percentage...
Electrical FLT-02	`data/electrical/ee-flt02.js#Q77`	Contractor buys equipment for Rs 240000 and sells for Rs 276000. Profit percent on cost is:

No local diagram files were resolved for this rule's sample set (text-only evidence). Figure-dependent rules should accumulate imaged bank items.

Elimination Rule 17 — Last-resort: dimensional / absurdity cull (evidence-limited)

Evidence: ESTABLISHED — 232 independent (231 bank + 1 VALID PYQ) · **Streams:** Civil, Electrical, Non-core

What this rule means

When two options remain, cull any survivor whose units cannot equal the asked quantity (mm^2 vs N/mm^2, ohm vs siemens, kN vs kn·m).

When to use

Time pressure; two options already killed by other established rules; two remain.

Application steps

1. Write the unit of the asked quantity.
2. Strike options whose units cannot match.
3. Then prefer consistency with figure boundary conditions — never 'pick C' myths.

```
Memory cue: Units(asked) must equal Units(option)
```

Core elimination move

Cull any remaining option whose units cannot match the ask (mm^2 vs N/mm^2, Ω vs S, kN vs kN·m). Then prefer the option consistent with the figure's boundary conditions. Do **not** guess from 'middle value' myths — our banks do not support that heuristic.

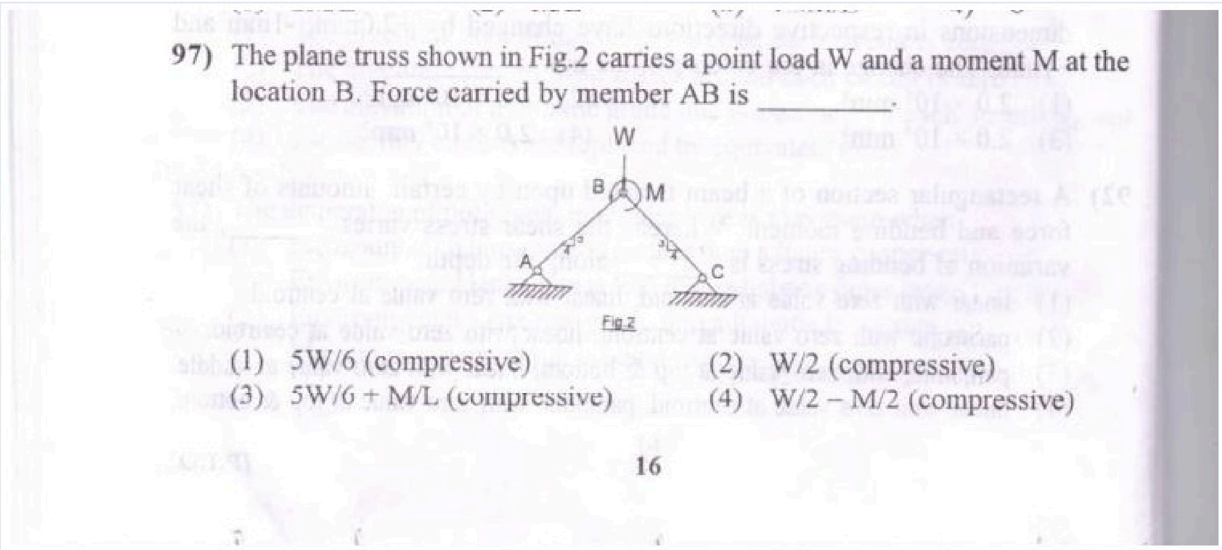
Evidence examples (sample references)

Source exam	Reference	Stem (truncated)
Civil FLT-01	`data/civil/ce-flt01.js#Q41`	The plane truss shown carries load W and moment M at apex B. Force in member AB is:
Civil FLT-02	`data/civil/ce-flt02.js#Q47`	The design buckling-curve crop plots reduction factor \(\chi\) against nondimensional slenderness for curve…
Civil ST-FE	`data/civil/st/ce-st-fe-01.js#Q1`	For a strip footing, take c = 20 kPa, Nc = 25, q = 30 kPa, Nq = 12, γ = 18 kN/m³, B = 2 m and Nγ = 8. Terza…
Civil ST-FM	`data/civil/st/ce-st-fm-01.js#Q1`	A mercury manometer connected to water shows a level difference of 0.4 m. Taking mercury specific gravity a…
Civil ST-RCC	`data/civil/st/ce-st-rcc-01.js#Q1`	An RCC beam of M25 and Fe500 has factored Mu = 180 kN m, b = 230 mm and d = 450 mm. Using z = 0.9d, the req…
Civil ST-SOIL	`data/civil/st/ce-st-soil-01.js#Q1`	A fine-grained soil has liquid limit 48% and plastic limit 22%. Its plasticity index is closest to:

Figure evidence (actual diagrams — not path titles)

Civil FLT-01 · `data/civil/ce-flt01.js#Q41` · Key A

The plane truss shown carries load W and moment M at apex B. Force in member AB is:



97) The plane truss shown in Fig.2 carries a point load W and a moment M at the location B. Force carried by member AB is ______.

(1) 5W/6 (compressive) (2) W/2 (compressive)
(3) 5W/6 + M/L (compressive) (4) W/2 – M/2 (compressive)

16

A. 5W/6 (compressive)

B. W/2 (compressive)

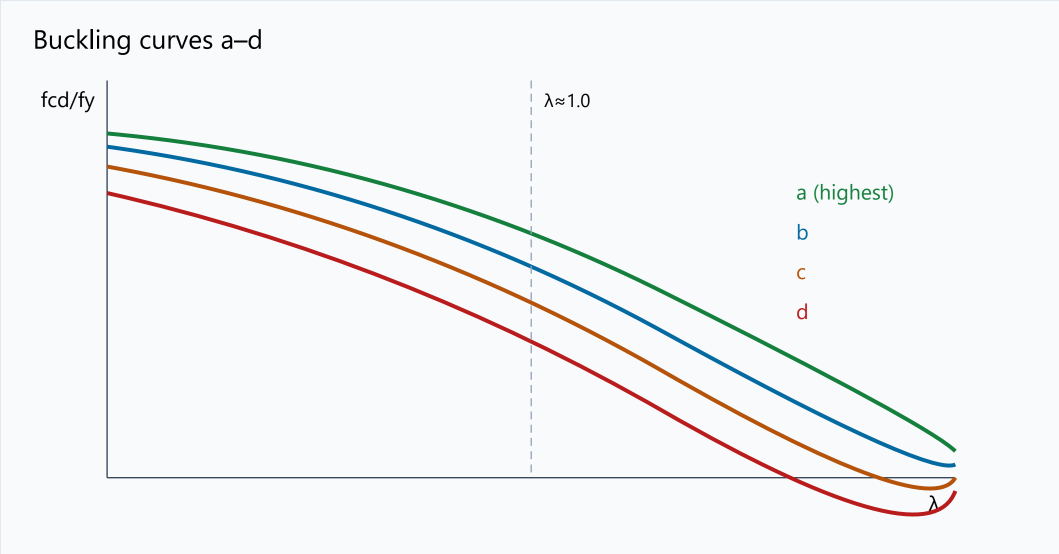
C. 5W/6 + M/L (compressive)

D. W/2 − M/2 (compressive)

Embedded figure file: images/diagrams/civil-flt01/q-extra-truss-ab.jpg

Civil FLT-02 · `data/civil/ce-flt02.js#Q47` · Key D

The design buckling-curve crop plots reduction factor \(\chi\) against nondimensional slenderness for curves a–d. A column's section/fabrication class and \(\bar\lambda\) are marked on the figure; read \(\chi\) and estim



A. 682 kN

B. 455 kN

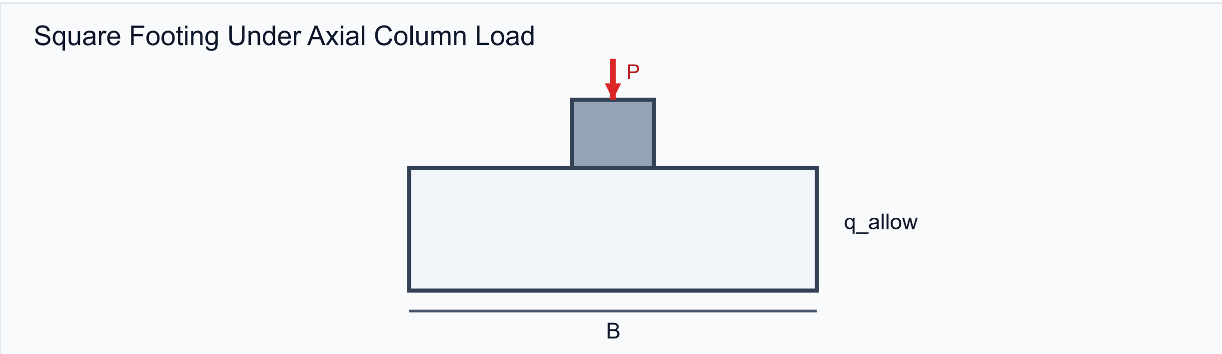
C. 364 kN

D. 545 kN

Embedded figure file: images/diagrams/civil-flt02/q47-buckling-curves.svg

Civil ST-FE · `data/civil/st/ce-st-fe-01.js#Q3` · Key D

The shown square footing distributes a concentric service load of 260 kN over a 2.3 m × 2.3 m base. The average contact pressure is closest to:



A. 39.32 kPa

B. 58.98 kPa

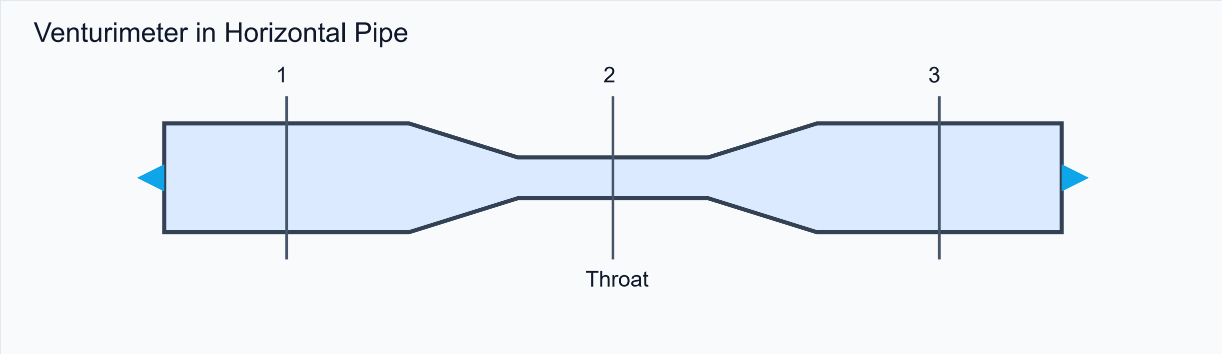
C. 73.72 kPa

D. 49.15 kPa

Embedded figure file: images/diagrams/civil-st-fe/foundation-square-footing.svg

Civil ST-FM · `data/civil/st/ce-st-fm-01.js#Q3` · Key D

The shown venturimeter is installed in a 0.2 m diameter inlet pipe. If the inlet velocity is 3 m/s, the discharge entering the convergent section is closest to:



A. 0.08 m^3/s

B. 0.11 m^3/s

C. 0.14 m^3/s

D. 0.09 m^3/s

Embedded figure file: images/diagrams/civil-st-fm/fluid-venturimeter.svg

3. Draft rules (not established — do not teach as heuristics)

These candidates have fewer than 15 independent hits. Keep them as a watchlist only.

- Kill the neighbour-formula lookalike — 4 hits
- SIL / surge-impedance identity — 5 hits
- Transformer referral / turns-ratio scaling — 7 hits

4. Anti-rules (unsupported myths)

- **Pick C when unsure** — keys are balanced; letter myths fail.
- **Longest option is correct** — distractors are equal-length engineering mistakes.
- **Always middle numerical value** — middle is often the classic slip (×2 / ÷2).
- **Ignore the figure if the stem looks complete** — imaged items are figure-dependent.

Generated for APTRANSCO AEE CBT practice. Regenerate: python scripts/build_elimination_playbook_pdf.py